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(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
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THE MARITIME SAFETY COMMITTEE,
RECALLING Article Z8(b) ofthe Convention on the International Maritime Organization concerning the functions of the Committee,
RECALLING ALSO resolution A.744(I8) by which the Assembly adopted the Guidelines on the enhanced programme of inspections during surveys of bulk carriers and oil tankers (the Guidelines),
RECALLING FURTHER article VIII(b) and regulation XI-l IZ of the International Convention for the Safety of Life at Sea (SOLAS), 1974 (hereinafter referred to as "the Convention") concerning the procedure for amending the Guidelines,
NOTING that the Assembly, when adopting resolution A.744(l8), requested the Maritime Safety Committee and the Marine Environment Protection Committee to keep the

Convention") concerning the procedure for amending the Guidelines,
NOTING that the Assembly, when adopting resolution A.744(l8), requested the Maritime Safety Committee and the Marine Environment Protection Committee to keep the Guidelines under review and update them as necessary, in the light of experience gained in their application,
NOTING ALSO resolutions MSC.49(66), MSC.I05(73), MSC.lZ5(75), MSC.144(77) and resolution Z of the 1997 Conference of Contracting Governments to the Convention, by which amendments to resolution A.744(l8) were adopted by the Maritime Safety Committee and the Conference of Contracting Governments to the Convention, in accordance with article VIII(b) and regulation XI-liZ of the Convention,
HAVING CONSIDERED, at its eightieth session, amendments to the Guidelines proposed and circulated in accordance with article VIII(b)(i) ofthe Convention,
I. ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the Guidelines on the enhanced programme of inspections during surveys of bulk carriers and oil tankers, the text of which is set out in the Annex to the present resolution;

Guidelines on the enhanced programme of inspections during surveys of bulk carriers and oil tankers, the text of which is set out in the Annex to the present resolution;
Z. DETERMINES, in accordance with article VIII(b)(vi)(Z)(bb) of the Convention, that the amendments shall be deemed to have been accepted on I July Z006, unless, prior to that date, more than one third of the Contracting Governments to the Convention or Contracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet, have notified their objections to the amendments;
3. INVITES Contracting Governments to note that, in accordance with article VIII(b)(vii)(Z) of the Convention, the amendments shall enter into force on 1 January Z007 upon their acceptance in accordance with paragraph Z above;
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4. REQUESTS the Secretary-General, in conformity with article VIII(b)(v) of the

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
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4. REQUESTS the Secretary-General, in conformity with article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the Annex to all Contracting Governments to the Convention;

5. FURTHER REQUESTS the Secretary-General to transmit copies of this resolution and its Annex to Members of the Organization, which are not Contracting Governments to the Convention.

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GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS

I The section "Contents" is replaced with the following and the relevant headings in the

(RESOLUTION A.744(18), AS AMENDED)
GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS
I The section "Contents" is replaced with the following and the relevant headings in the text of the Guidelines are amended accordingly:
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Annex 11 Guidelines for the gauging of the vertically corrugated transverse watertight bulkhead between holds Nos.l and 2
Annex 12 Additional annual survey requirements for the foremost cargo hold of ships subject to SOLAS regulation XIII9.l
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2 Throughout the Guidelines:
.1 the words "enhanced survey during the periodical survey", "periodical survey" and "enhanced survey" are replaced with the words "renewal survey";
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.2 the words "enhanced survey during annual survey" are replaced with the words "annual survey";
.3 the words "intermediate enhanced survey" are replaced with the words "intermediate survey"; and
.4 the words "is to be" and "are to be" are replaced with the words "should be".
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GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
3 A new paragraph 1.1.1 is added as follows:
" 1.1.1 The Guidelines should apply to all self-propelled bulk carners of 500 gross tonnage and above."
Existing paragraphs 1.1.1 and 1.1.2 are renumbered as paragraphs 1.1.2 and 1.1.3.
4 In new paragraph 1.1.2 (existing paragraph 1.1.1), in the second sentence, the words "regulation 1110 of' are inserted between "by" and "the 1974 SOLAS Convention".
5 In paragraph 1.2.14, the words "Intermediate enhanced survey is an enhanced survey" are replaced by "Intermediate survey is a survey".

"regulation 1110 of' are inserted between "by" and "the 1974 SOLAS Convention".
5 In paragraph 1.2.14, the words "Intermediate enhanced survey is an enhanced survey" are replaced by "Intermediate survey is a survey".
6 In existing paragraph 1.2.15, the words "condition of classification" are replaced by "condition of classification or recommendation".
7 A new paragraph 1.2.17 is added as follows:
"1.2.17 Specially considered means sufficient close-up inspection and thickness measurements are taken to confirm the actual average condition of the structure under coating."
8 The word "significant" in paragraph 1.3.2 is deleted.
9 The following new paragraph 1.4 is added:
"1.4 Surveyors
For bulk carriers of 20,000 tons deadweight and above, two surveyors should jointly carry out the first scheduled renewal survey after the bulk carrier passes 10 years of age, and all subsequent renewal surveys and intermediate surveys. If the surveys are carried out by a recognized organization, the surveyors should be exclusively employed by such recognized organizations."
10 The words "thickness measurement and" in paragraph 2.1.2 are deleted.
II Paragraph 2.2.4 is deleted.

recognized organization, the surveyors should be exclusively employed by such recognized organizations."

10 The words "thickness measurement and" in paragraph 2.1.2 are deleted.

II Paragraph 2.2.4 is deleted.

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12 The references to a footnote regarding "specially considered" in paragraphs 2.3.1, 2.6.4, 3.4.1.1 and 3.4.2.1 are deleted.

13 In paragraph 3.1, the word "annual" is inserted before the word "survey".

14 Existing paragraphs 3.3.2, 3.3.4,3.3.5 and 3.3.6 are deleted.

15 The following new paragraphs 3.3.2 to 3.3.7 are added after existing paragraph 3.3.1:

"3.3.2 A thorough survey of cargo hatch covers and coamings is only possible by examination in the open as well as closed positions and should include verification of proper opening and closing operation. As a result, the hatch cover sets within the forward 25% of the ship's length and at least one additional set, such that all sets on the ship are

proper opening and closing operation. As a result, the hatch cover sets within the forward 25% of the ship's length and at least one additional set, such that all sets on the ship are assessed at least once in every 5-year period, should be surveyed open, closed and in operation to the full extent in each direction at each annual survey, including:

.1 stowage and securing in open condition;

.2 proper fit and efficiency of sealing in closed condition; and

.3 operational testing of hydraulic and power components, wires, chains and link drives.

The closing of the covers should include the fastening of all peripheral, and cross joint cleats or other securing devices. Particular attention should be paid to the condition of hatch covers in the forward 25% of the ship's length, where sea loads are normally greatest.

3.3.3 If there are indications of difficulty in operating and securing hatch covers, additional sets above those required by 3.3.2, at the discretion of the surveyor, should be tested in operation.

3.3.4 Where the cargo hatch securing system does not function properly, repairs should

additional sets above those required by 3.3.2, at the discretion of the surveyor, should be tested in operation.

3.3.4 Where the cargo hatch securing system does not function properly, repairs should be carried out under the supervision of the Administration. Where hatch covers or coamings undergo substantial repairs, the strength of securing devices should be upgraded to comply with annex 13.

3.3.5 For each cargo hatch cover set, at each annual survey, the following items should be surveyed:

.1 cover panels, including side plates, and stiffener attachments that may be accessible in the open position by close-up survey (for corrosion, cracks, deformation);

.2 sealing arrangements of perimeter and cross joints (gaskets for condition and permanent deformation, flexible seals on combination carriers, gasket lips, compression bars, drainage channels and non return valves);

.3 clamping devices, retaining bars, cleating (for wastage, adjustment, and condition of rubber components);

.4 closed cover locating devices (for distortion and attachment);

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condition of rubber components);
.4 closed cover locating devices (for distortion and attachment);
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.5 chain or rope pulleys;
.6 guides;
.7 guide rails and track wheels;
.8 stoppers;
.9 wires, chains, tensioners and gypsies;
.10 hydraulic system, electrical safety devices and interlocks; and
. I I end and interpanel hinges, pins and stools where fitted.
3.3.6 At each hatchway, at each annual survey, the coamings, with plating, stiffeners and brackets should be checked for corrosion, cracks and deformation, especially of the coaming tops.
3.3.7 Where considered necessary, the effectiveness of sealing arrangements may be proved by hose or chalk testing supplemented by dimensional measurements of seal compressing components."
16 Existing paragraph 3.3.3 is renumbered as 3.3.8.
I 7 Existing paragraph 5 .1. I is replaced by the following:
"5.1.1 A specific survey programme should be worked out in advance of the renewal

compressing components."
16 Existing paragraph 3.3.3 is renumbered as 3.3.8.
I 7 Existing paragraph 5 .1. I is replaced by the following:
"5.1.1 A specific survey programme should be worked out in advance of the renewal survey by the owner in co-operation with the Administration. The survey programme should be in a written format based on the information in annex 4A. The survey should not commence until the survey programme has been agreed.
5.1.1.1 Prior to the development of the survey programme, the survey planning questionnaire should be completed by the owner based on the information set out in annex 4B, and forwarded to the Administration."
18 The seven subitems in paragraph 5.1.2 are numbered from ".1" to ".7" and the eleven subitems in paragraph 5.1.3 are numbered from ".1" to ".II".
19 Paragraph 5.1.4 is deleted and paragraphs 5.1.5 and 5.1.6 are renumbered as 5.1.4 and 5.1.5.
20 The following new paragraphs 5.2.1.1 to 5.2.1.3 are added:
"5.2.1.1 In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access should be agreed between the owner and the Administration.

20 The following new paragraphs 5.2.1.1 to 5.2.1.3 are added:
"5.2.1.1 In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access should be agreed between the owner and the Administration.
5.2.1.2 Details of the means of access should be provided in the survey planning questionnaire.
5.2.1.3 In cases where the provisions of safety and required access are judged by the attending surveyors not to be adequate, the survey of the spaces involved should not proceed."
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21 Existing paragraphs 5.2.2,5.2.3 and 5.2.4 are replaced by the following:
"5.2.2 Cargo holds, tanks and spaces should be safe for access. Cargo holds, tanks and spaces should be gas free and properly ventilated. Prior to entering a tank, void or enclosed space, it should be verified that the atmosphere in the tank is free from hazardous gas and contains sufficient oxygen.

spaces should be gas free and properly ventilated. Prior to entering a tank, void or enclosed space, it should be verified that the atmosphere in the tank is free from hazardous gas and contains sufficient oxygen.

5.2.3 Cargo holds, tanks and spaces should be sufficiently clean and free from water, scale, dirt, oil residues, sediments, etc., to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating. In particular this applies to areas which are subject to thickness measurement.

5.2.4 Sufficient illumination should be provided to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating."

22 The following new paragraphs 5.2.5 and 5.2.6 are added:

"5.2.5 The surveyor(s) should always be accompanied by at least one responsible person, assigned by the owner, experienced in tank and enclosed spaces inspection. In addition a back-up team of at least two experienced persons should be stationed at the hatch opening of the tank or space that is being surveyed. The back -up team should continuously

back-up team of at least two experienced persons should be stationed at the hatch opening of the tank or space that is being surveyed. The back -up team should continuously observe the work in the tank or space and should keep life-saving and evacuation equipment ready for use.

5.2.6 A communication system should be arranged between the survey party in the cargo hold, tank or space being examined, the responsible officer on deck and, as the case may be, the navigation bridge. The communication arrangements should be maintained throughout the survey."

23 In existing paragraph 5.3.2, a new subitem "portable ladders" is inserted between the third and fourth subitems.

24 The five subitems in paragraph 5.3.2 are numbered from ".1" to ".5" and the five subitems in paragraph 5.4.2 are numbered from ".1" to ".5".

25 New paragraphs 5.4.3 to 5.4.5 are added as follows:

"5.4.3 Explosimeter, oxygen-meter, breathing apparatus, lifelines, riding belts with rope and hook and whistles together with instructions and guidance on their use should be made available during the survey. A safety checklist should be provided.

and hook and whistles together with instructions and guidance on their use should be made available during the survey. A safety checklist should be provided.
5.4.4 Adequate and safe lighting should be provided for the safe and efficient conduct ofthe survey.
5.4.5 Adequate protective clothing should be made available and used (e.g. safety helmet, gloves, safety shoes, etc.) during the survey."
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26 The existing paragraph 5.5.3 is replaced by the following:
"5.5.3 When rafts or boats will be used for close-up survey the following conditions should be observed:
.1 only rough duty, inflatable rafts or boats, having satisfactory residual buoyancy and stability even if one chamber is ruptured, should be used;
.2 the boat or raft should be tethered to the access ladder and an additional person should be stationed down the access ladder with a clear view of the boat or raft;
.3 appropriate lifejackets should be available for all participants;

.2 the boat or raft should be tethered to the access ladder and an additional person should be stationed down the access ladder with a clear view of the boat or raft;
.3 appropriate lifejackets should be available for all participants;
.4 the surface of water in the tank or hold should be calm (under all foreseeable conditions the expected rise of water within the tank should not exceed 0.25 m) and the water level either stationary or falling. On no account should the level of the water be rising while the boat or raft is III use;
.5 the tank, hold or space must contain clean ballast water only. Even a thin sheen of oil on the water is not acceptable; and
.6 at no time should the water level be allowed to be within I m of the deepest under-deck web face flat so that the survey team is not isolated from a direct escape route to the tank hatch. Filling to levels above the deck transverses should only be contemplated if a deck access manhole is fitted and open in the bay being examined, so that an escape route for the survey party is available at all times. Other effective means of escape to the deck may be considered."

fitted and open in the bay being examined, so that an escape route for the survey party is available at all times. Other effective means of escape to the deck may be considered."

27 The following new paragraphs 5.5.4 to 5.5.6 are added:

"5.5.4 Rafts or boats alone may be allowed for inspection of the under-deck areas for tanks or spaces, if the depth of the webs is 1.5 m or less.

5.5.5 If the depth of the webs is more than 1.5 m, rafts or boats alone may be allowed only:

.I when the coating of the under-deck structure is in GOOD condition and there is no evidence of wastage; or

.2 if a permanent means of access is provided in each bay to allow safe entry and exit. This means of access should be direct from the deck via a vertical ladder with a small platform fitted approximately 2 m below the deck. Other effective means of escape to the deck may be considered.

If neither of the above conditions are met, then staging or other equivalent means should be provided for the survey of the under-deck areas.

5.5.6 The use of rafts or boats alone in paragraphs 5.5.4 and 5.5.5 does not preclude the use of boats or rafts to move about within a tank during a survey."

means should be provided for the survey of the under-deck areas.
5.5.6 The use of rafts or boats alone in paragraphs 5.5.4 and 5.5.5 does not preclude the use of boats or rafts to move about within a tank during a survey."
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 14 -
28 The following new section 5.6 is added:
"5.6 Survey planning meeting
5.6.1 The establishment of proper preparation and the close co-operation between the attending surveyor(s) and the owner's representatives onboard prior to and during the survey are an essential part in the safe and efficient conduct of the survey. During the survey on board safety meetings should be held regularly.
5.6.2 Prior to commencement of any part of the renewal and intermediate survey, a survey planning meeting should be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement company operator (as applicable) and the master of the ship for the purpose to ascertain that all the arrangements envisaged

representative in attendance, the thickness measurement company operator (as applicable) and the master of the ship for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

5.6.3 The following is an indicative list of items that should be addressed 10 the meeting:

.1 schedule of the vessel (i.e. the voyage, docking and undocking manoeuvres, periods alongside, cargo and ballast operations, etc.);

.2 provisions and arrangements for thickness measurements (i.e. access, cleaning/de-scaling, illumination, ventilation, personal safety);

.3 extent of the thickness measurements ;

.4 acceptance criteria (refer to the list of minimum thicknesses);

.5 extent of close-up survey and thickness measurement considering the coating condition and suspect areas/areas of substantial corrosion;

.6 execution of thickness measurements;

.7 taking representative readings ill general and where uneven corrosion/pitting is found;

.8 mapping of areas of substantial corrosion; and

.9 communication between attending surveyor(s) the thickness measurement

.7 taking representative readings ill general and where uneven
corrosion/pitting is found;
.8 mapping of areas of substantial corrosion; and
.9 communication between attending surveyor(s) the thickness measurement
company operator(s) and owner representative(s) concerning findings."
29 The words "supply and maintain on-board" in paragraph 6.1.1 are replaced with "obtain,
supply and maintain on board the ship".
30 Paragraph 6.2.1.4 is deleted.
31 The word "inspection" in paragraph 6.4 is replaced with "survey".
32 The word "extend" in paragraph 7.1.3 is replaced with "extent".
33 The following new paragraph 8.2.2 is added after existing paragraph 8.2.1:
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 15 -
"8.2.2 When a survey is split between different survey stations, a report should be made
for each portion of the survey. A list of items examined andlor tested (pressure testing,
thickness measurements, etc.) and an indication of whether the item has been credited,

for each portion of the survey. A list of items examined andlor tested (pressure testing, thickness measurements, etc.) and an indication of whether the item has been credited, should be made available to the next attending surveyor(s), prior to continuing or completing the survey."

34 The existing paragraph 8.2.2 is renumbered as 8.2.3.

35 The existing annex 4 is deleted.

36 The following new annex 4A is added after existing annex 3:

"ANNEX4A

SURVEY PROGRAMME

Basic information and particulars

Name of ship:

IMO number:

Flag State:

Port of registry :

Gross tonnage:

Deadweight (metric tonnes) :

Length between perpendiculars (m) :

Shipbuilder:

Hull number:

Recognized organization (RO) :

RO ship identity:

Date of delivery of the ship:

Owner:

Thickness measurement company:

1 Preamble

1.1 Scope

1.1.1 The present survey programme covers the minimum extent of overall surveys, close-up surveys, thickness measurements and pressure testing within the cargo length area, cargo holds, ballast tanks, including fore and aft peak tanks, required by the Guidelines.

1.1.2 The arrangements and safety aspects of the survey should be acceptable to the

area, cargo holds, ballast tanks, including fore and aft peak tanks, required by the Guidelines.
1.1.2 The arrangements and safety aspects of the survey should be acceptable to the attending surveyor(s).
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 16 -
1.2 Documentation
All documents used in the development of the survey programme should be available on board during the survey as required by section 6.
2 Arrangement of cargo holds, tanks and spaces
This section of the survey programme should provide information (either in the form of plans or text) on the arrangement of cargo holds, tanks and spaces that fall within the scope ofthe survey.
3 List of cargo holds, tanks and spaces with information on their use, extent of coatings and corrosion protection system
This section of the survey programme should indicate any changes relating to (and should update) the information on the use of the holds and tanks of the ship, the extent of coatings and the corrosion protective system provided in the Survey Planning

update) the information on the use of the holds and tanks of the ship, the extent of coatings and the corrosion protective system provided in the Survey Planning Questionnaire.

4 Conditions for survey

This section of the survey programme should provide information on the conditions for survey, e.g. information regarding cargo hold and tank cleaning, gas freeing, ventilation, lighting, etc.

5 Provisions and method of access to structures

This section of the survey programme should indicate any changes relating to (and should update) the information on the provisions and methods of access to structures provided in the Survey Planning Questionnaire.

6 List of equipment for survey

This section of the survey programme should identifY and list the equipment that will be made available for carrying out the survey and the required thickness measurements .

7 Survey requirements

7.1 Overall survey

This section of the survey programme should identifY and list the spaces that should undergo an overall survey for this ship in accordance with 2.4.1 and 2.5.1.

7.2 Close-up survey

This section of the survey programme should identify and list the hull structures that

undergo an overall survey for this ship in accordance with 2.4.1 and 2.5.1.
7.2 Close-up survey
This section of the survey programme should identify and list the hull structures that should undergo a close-up survey for this ship in accordance with 2.5.2.
8 Identification of tanks for tank testing
This section of the survey programme should identifY and list the cargo holds and tanks that should undergo tank testing for this ship in accordance with 2.7.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 17 -
9 Identification of areas and sections for thickness measurements
This section of the survey programme should identify and list the areas and sections where thickness measurements should be taken in accordance with 2.6.1.
10 Minimum thickness of hull structures
This section of the survey programme should specify the minimum thickness for hull structures of this ship that are subject to survey, according to (a) or (b):
(a) D Determined from the attached wastage allowance table and the original

This section of the survey programme should specify the minimum thickness for hull structures of this ship that are subject to survey, according to (a) or (b):
(a) D Determined from the attached wastage allowance table and the original thickness to the hull structure plans of the sbip ;
(b) D Given in the following table(s):
Area or location Original as-built Minimum Substantial thickness thickness corrosion (mm) (mm) thickness (mm)
Deck
Plating
Longitudinals
Longitudinal girders
Cross deck olating
Cross deck stiffeners
Bottom
Platin!!
Lon"itudinals
Longitudinal girders
Inner bottom
Plating
Longitudinals
Lon!!itudinal girders
Floors
Ship side in way of topside tanks
Platin!!
Lon!!itudinals
Ship side in way of hopper side tanks
Platin!!
Longitudinals
Ship side in way of tanks (if annJicable)
Platin!!
Longitudinals
Longitudinal stringers
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 18 -
Ship side in way of cargo holds
Plating
Side frames webs
Side frames flanges
Upper brackets webs

PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 18 -

Ship side in way of cargo holds
Plating
Side frames webs
Side frames flanges
Upper brackets webs
Upper brackets flanges
Lower brackets webs
Lower brackets flanges
Longitndinal bulkhead if applicable
Plating
Longitudinals, if applicable
Longitudinal girders, if applicable
Transverse bulkheads
Plating
Stiffeners, if applicable
Upper stool plating
Upper stool stiffeners
Lower stool plating
Lower stool stiffeners
Transverse web frames in topside tanks
Plating
Flanges
Stiffeners
Transverse web frames in hopper tanks
Plating
Flanges
Stiffeners
Hatch covers
Plating
Stiffeners
Hatch coamings
Plating
Stiffeners

Note: The wastage allowance tables should be attached to the survey programme.

11 Thickness measurement company

This section of the survey programme should identify changes, if any, relating to the information on the thickness measurement company provided in the Survey Planning Questionnaire.

12 Damage experience related to the ship

This section of the survey programme should, using the tables provided below, provide

information on the thickness measurement company provided in the Survey Planning Questionnaire.

12 Damage experience related to the ship

This section of the survey programme should, using the tables provided below, provide details of the hull damages for at least the last three years in way of the cargo holds, ballast tanks and void spaces within the cargo length area. These damages are subject to survey.

RESOLUTION MSC.197(80)

(adopted on 20 May 2005)

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)Cargo hold, tank or space number or area

Cargo hold, tank or space number or area

Hull damages sorted by location for this ship

Possible cause, Description Location Repair if known of the damages

Hull damages for sister or similar ships (if available) in the case of design related damage

Possible cause, Description Location Repair if known of the damages

13 Areas identified with substantial corrosion from previous surveys

Date of repair

Date of repair

This section of the survey programme should identifY and list the areas of substantial

if known of
the damages
13 Areas identified with substantial corrosion from previous surveys
Date of
repair
Date of
repair
This section of the survey programme should identifY and list the areas of substantial corrosion from previous surveys.
14 Critical structural areas and suspect areas
This section of the survey programme should identify and list the critical structural areas and the suspect areas, when such information is available.
15 Other relevant comments and information
This section of the survey programme should provide any other comments and information relevant to the survey.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 20-
Appendices
Appendix 1 - List of plans
Paragraph 5.1.3.2 requires that main structural plans of cargo holds and ballast tanks (scantling drawings), including information on regarding use of high tensile steel (HTS) should be available. This Appendix of the survey programme should identify and list the main structural plans which form part of the survey programme .

(scantling drawings), including information on regarding use of high tensile steel (HTS) should be available. This Appendix of the survey programme should identify and list the main structural plans which form part of the survey programme .

Appendix 2 - Survey Planning Questionnaire

The Survey Planning Questionnaire (annex 4B), which has been submitted by the owner, should be appended to the survey programme .

Appendix 3 - Other documentation

This part of the survey programme should identify and list any other documentation that forms part of the plan.

Prepared by the owner in co-operation with the Administration for compliance with 5.1.3:

Date: ....

Date: ....

(name and signature of authorized owner's representative)

(name and signature of authorized r epresentative of the Administration)"

37 The following new Annex 4B is added after Annex 4A:

"ANNEX4B

SURVEY PLANNING QUESTIONNAIRE

I The following information will enable the owner in co-operation with the Administration to develop a Survey Plan complying with the requirements of the

"ANNEX4B
SURVEY PLANNING QUESTIONNAIRE
I The following information will enable the owner in co-operation with the Administration to develop a Survey Plan complying with the requirements of the Guidelines. It is essential that the owner provides, when completing the present questionnaire, up-to-date information. The present questionnaire, when completed, should provide all information and material required by the Guidelines.
Particulars
Ship's name:
IMO number:
Flag State:
Port of registry:
Owner :
Recognized organization:
Gross tonnage:
Deadweight (metric tonnes):
Date of delivery:
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 21 -
Information on access provision for close-up surveys and thickness measurement
2 The owner should indicate, in the table below, the means of access to the structures subject to close-up survey and thickness measurement. A close-up survey is an examination where the details of structural components are within the close visual

structures subject to close-up survey and thickness measurement. A close-up survey is an examination where the details of structural components are within the close visual inspection range of the attending surveyor, i.e. preferably within reach ofhand.

Holdrrank Structure Temporary Rafts Ladders Direct Other
No. staging access means
(please
specify)
F.P. Fore peak
A.P. Aft peak
Hatch side coarnings
Topside s loping plate
Upper stool plating
~
Cross deck :2
0
Side shell, frarnes and -"
0 brackets en
til Transverse bulkhead U
Hopper tank platting
Lower stool
Tank top
"
Under-deck structure
-0 ~
Side shell and structure .- -"
~ " §'.s Sloping plate and structure
E-< Webs and bulkheads
Hopper sloping plate and
~
structure
§:~ Side shell and structure o " ;x:- Bottom structure
Webs and bulkheads
Double bottom structure
Upper stool internal
structure
Lower stool internal
structure
Under-deck and structure
Side shell and structure
~ Side shell vertical web and
4-0 ~
o 'E structure
~ " Longitudinal bulkhead and ~ <>
" " structure - ~ eno Longitudinal bulkhead web ,,~
. - .D
::: ::s and structure 0
Bottom plating and -0
structure

Side shell and structure
~ Side shell vertical web and
4-0 ~
o 'E structure
~ " Longitudinal bulkhead and ~ <>
" " structure - ~ eno Longitudinal bulkhead web ,,~
. - .D
::: ::s and structure 0
Bottom plating and -0
structure
Cross lies/stringers
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 22-
History of bulk cargoes of a corrosive natnre (e.g. high sulphur content)
Owner's inspections
3 Using a fonnat similar to that of the table below (which is given as an example),
the owner should provide details of the results of their inspections, for the last 3 years - in
accordance with the Guidelines - on all CARGO holds and BALLAST tanks and VOID
spaces within the cargo area.
TankIHold Corrosion Coating
No. protection extent
(1) (2)
Cargo holds
Topside tanks
Hopper tanks
Double side
skin tanks
Double
bottom tanks
Upper stools
Lower stools
Wing tanks
(ore tankers)
Fore peak
Aft peak
Miscellaneous
other spaces:
Note: Indicate tanks which are used for oiVballast

Cargo holds
Topside tanks
Hopper tanks
Double side
skin tanks
Double
bottom tanks
Upper stools
Lower stools
Wing tanks
(ore tankers)
Fore peak
Aft peak
Miscellaneous
other spaces:
Note: Indicate tanks which are used for oiVballast
1) HC=hard coating; SC=soft coating;
A=anodes; NP=no protection
2) U=upper part; M=middle part;
L=lower part; C=complete
3) G=good; F=fair; P=poor;
RC=recoated (during the last 3 years)
4) N=no findings recorded; Y=findings
recorded, description of findings should
be attached to this questionnaire
5) DR=Damage and repair; L=Leakages;
CV= Conversion (description to be
attached to this questionnaire)
Coating Struct ural Hold and
condition deterioration tank
(3) (4) history
(5)
Name of owner's representative:
Signature: ..... .
Date: ....
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 23 -
Reports of Port State Control inspections

(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 23 -
Reports of Port State Control inspections
List the reports of Port State Control inspections containing hull structural-related deficiencies, relevant information on rectification of the deficiencies:
Safety Management System
List non-conformities related to hull maintenance, including the associated corrective actions:
Name and address of the approved thickness measurement company:
"

38 The existing text of Annex 6 is replaced by the following:

"ANNEX 6
SURVEY REPORTING PRINCIPLES

As a principle, for bulk carriers subject to the Guidelines, the surveyor should include the following contents in his report for survey of hull structure and piping systems, as relevant for the survey.

1 General

l.l A survey report should be generated in the following cases:

.1 in connection with commencement, continuation and/or completion of periodical hull surveys, i.e. annual, intermediate and renewal surveys, as relevant;

.2 when structural damages/defects have been found;

.1 in connection with commencement, continuation and/or completion of periodical hull surveys, i.e. annual, intermediate and renewal surveys, as relevant;
.2 when structural damages/defects have been found;
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 24-
.3 when repairs, renewals or modifications have been carried out; and
.4 when condition of class (recommendation) has been imposed or has been deleted.
1.2 The reporting should provide:
.1 evidence that prescribed surveys have been carried out in accordance with applicable requirements;
.2 documentation of surveys carried out with findings, repairs carried out and condition of class (recommendation) imposed or deleted;
.3 survey records, including actions taken, which should form an auditable documentary trail. Survey reports should be kept in the survey report file required to be on board;
.4 information for planning of future surveys; and
.5 information which may be used as input for maintenance of classification rules and instructions.

required to be on board;
.4 information for planning of future surveys; and
.5 information which may be used as input for maintenance of classification rules and instructions.
1.3 When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items surveyed, relevant findings and an indication of whether the item has been credited, should be made available to the next attending surveyor, prior to continuing or completing the survey. Thickness measurement and tank testing carried out is also to be listed for the next surveyor.
2 Extent of the survey
2.1 Identification of compartments where an overall survey has been carried out.
2.2 Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where a close-up survey has been carried out, together with information on the means of access used.
2.3 Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where thickness measurement has been carried out.
Note: As a minimum, the identification of location of close-up survey and thickness

2.3 Identification of locations, in each ballast tank and cargo hold including hatch covers and coamings, where thickness measurement has been carried out.
Note: As a minimum, the identification of location of close-up survey and thickness measurement should include a confirmation with description of individual structural members corresponding to the extent of requirements stipulated in Annex A based on type of periodical survey and the ship's age.
Where only partial survey is required, i.e. 25% of shell frames, one transverse web, two selected cargo hold transverse bulkheads, the identification should include location within each ballast tank and cargo hold by reference to frame numbers.
2.4 For areas in ballast tanks and cargo holds where protective coating is found to be in good condition and the extent of close-up survey and/or thickness measurement has been specially considered, structures subject to special consideration should be identified.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 25 -

RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 25 -
2.5 Identification of tanks subject to tank testing.
2.6 Identification of piping systems on deck and within cargo holds , ballast tanks , pipe tunnels, cofferdams and void spaces where:
.1 examination including internal examination of piping with valves and fittings and thickness measurement , as relevant , has been carried out; and
.2 operational test to working pressure has been carried out.
3 Result of the survey
3.1 Type, extent and condition of protective coating in each tank , as relevant (rated GOOD , FAIR or POOR) including identification of tanks fitted with anodes.
3.2 Structural condition of each compartment with information on the following , as relevant:
.1 identification of findings , such as:
.1.1 corrosion with description of location, type and extent;
.1.2 areas with substantial corrosion;
.1.3 cracks /fractures with description oflocation and extent;
.1.4 buckling with description of location and extent; and

.1.1 corrosion with description of location, type and extent;
.1.2 areas with substantial corrosion;
.1.3 cracks /fractures with description oflocation and extent;
.1.4 buckling with description of location and extent; and
.1.5 indents with description of location and extent;
.2 identification of compartments where no structural damages /defects are found. The report may be supplemented by sketches/photos; and
.3 thickness measurement report should be verified and signed by the surveyor controlling the measurements on board.
4 Actions taken with respect to findings
4.1 Whenever the attending surveyor is of the opinion that repair s are required , each item to be repaired should be identified in a numbered list. Whenever repairs are carried out, details of the repairs effected should be reported by making specific reference to relevant items in the numbered list.
4.2 Repairs carried out should be reported with identification of:
.1 compartment;
.2 structural member;
.3 repair method (i.e. renewal or modification) , including:
.3.1 steel grades and scantlings (if different from the original);
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)

.1 compartment;
.2 structural member;
.3 repair method (i.e. renewal or modification) , including:
.3.1 steel grades and scantlings (if different from the original);
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 26-
.3.2 sketches/photos, as appropriate;
.4 repair extent; and
.5 non-destructive test (NDT)/tests.
4.3 For repairs not completed at the time of survey, condition of class/recommendation should be imposed with a specific time limit for the repairs. In order to provide correct and proper information to the surveyor attending for survey of the repairs, condition of class/recommendation should be sufficiently detailed with identification of each item to be repaired. For identification of extensive repairs, reference may be made to the survey report."
39 Table 1 and table 2 in appendix 3 of Annex 8 are deleted and table 3 is renumbered as table I.
40 In paragraph I of annex 9, the words "the paragraph 5.1.6 of annex A" are replaced with "5.1.5".
41 The following new Annex 13 is added:
"ANNEX 13

table I.
40 In paragraph I of annex 9, the words "the paragraph 5.1.6 of annex A" are replaced with "5.1.5".
41 The following new Annex 13 is added:
"ANNEX 13
STRENGTH OF CARGO HATCH COVER SECURING ARRANGEMENTS FOR BULK CARRIERS
1 Securing devices
The strength of securing devices should comply with the following requirements:
.1 Panel hatch covers should be secured by appropriate devices (bolts, wedges or similar) suitably spaced alongside the coamings and between cover elements. Arrangement and spacing should be determined with due attention to the effectiveness for weathertightness, depending upon the type and the size of the hatch cover, as well as on the stiffness of the cover edges between the securing devices .
. 2 The net sectional area of each securing device is not to be less than:
where:
a - spacing between securing devices not to be taken less than 2 metres
f = (cry / 235)"
cry = specified minimum upper yield stress in N/mm2 of the steel used for fabrication, not to be taken greater than 70% of the ultimate tensile strength
e 0.75 for cry > 235
1.0 for cry :S 235
Rods or bolts should have a net diameter not less than 19 mm for

used for fabrication, not to be taken greater than 70% of the
ultimate tensile strength
e 0.75 for cry > 235
1.0 for cry :S 235
Rods or bolts should have a net diameter not less than 19 mm for
hatchways exceeding 5 m 2 in area.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 27-
.3 Between cover and coaming and at cross-joints, a packing line pressure
sufficient to obtain weathertightness should be maintained by the securing
devices. For packing line pressures exceeding 5 N/mm, the cross section
area should be increased in direct proportion. The packing line pressure
should be specified .
.4 The cover edge stiffness should be sufficient to maintain adequate sealing
pressure between securing devices. The moment of inertia, I, of edge
elements be less than:
where:
p packing line pressure in N/mm, minimum 5 N/mm
a = spacing in m of securing devices
.5 Securing devices should be of reliable construction and securely attached
to the hatchway coarnings, decks or covers. Individual securing devices

where:
p packing line pressure in N/mm, minimum 5 N/mm
a = spacing in m of securing devices
.5 Securing devices should be of reliable construction and securely attached to the hatchway coarnings, decks or covers. Individual securing devices on each cover are to have approximately the same stiffness characteristics .
. 6 Where rod cleats are fitted, resilient washers or cushions should be incorporated .
. 7 Where hydraulic cleating is adopted, a positive means should be provided to ensure that it remains mechanically locked in the closed position in the event offailure of the hydraulic system.
2 Stoppers
2.1 Nos.l and 2 hatch covers should be effectively secured, by means of stoppers, against the transverse forces arising from a pressure of 175 kN/m 2
2.2 No.2 hatch covers should be effectively secured, by means of stoppers, against the longitudinal forces acting on the forward end arising from a pressure of 175 kN/m2
2.3 No.1 hatch cover should be effectively secured, by means of stoppers, against the longitudinal forces acting on the forward end arising from a pressure of 230 kN/m2 This pressure may be reduced to 175 kN/m2 if a forecastle is fitted.

2.3 No.1 hatch cover should be effectively secured, by means of stoppers, against the longitudinal forces acting on the forward end arising from a pressure of 230 kN/m2 This pressure may be reduced to 175 kN/m2 if a forecastle is fitted.
2.4 The equivalent stress in stoppers and their supporting structures and calculated in the throat of the stopper welds is not to exceed the allowable value of 0.8 (Jy.
3 Materials and welding
Where stoppers or securing devices are fitted to comply with this annex, they should be manufactured of materials, including welding electrodes, to the satisfaction of the Administration. "
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
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ANNEXB
GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF OIL TANKERS
42 The text of existing Annex B is replaced by a new part A with the title:
"Part A
GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF DOUBLE HULL OIL TANKERS "
and a new part B with the title:
"Part B

42 The text of existing Annex B is replaced by a new part A with the title:
"Part A
GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF DOUBLE HULL OIL TANKERS "
and a new part B with the title:
"Part B
GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF OIL TANKERS OTHER THAN DOUBLE HULL OIL TANKERS"
43 The text of the new part A is as follows:
"Part A
GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF DOUBLE HULL OIL TANKERS
1 General
1.1 Application
1.1.1 The Guidelines should apply to all self-propelled double hull oil tankers of 500 gross tonnage and above.
1.1.2 The Guidelines should apply to surveys of hull structure and piping systems in way of cargo tanks, pump-rooms, cofferdams, pipe tunnels, void spaces within the cargo area and all ballast tanks. The surveys should be carried out during the surveys prescribed by regulation 1/10 of the 1974 SOLAS Convention, as amended.
1.1.3 The Guidelines contain the extent of examination, thickness measurements and tank testing. The survey should be extended when substantial corrosion and/or structural

prescribed by regulation 1/10 of the 1974 SOLAS Convention, as amended.
1.1.3 The Guidelines contain the extent of examination, thickness measurements and tank testing. The survey should be extended when substantial corrosion and/or structural defects are found and include additional close-up survey when necessary.
1.2 Definitions
1.2.1 Double hull oil tanker is a ship which is constructed primarily for the carriage of oil in bulk, which have the cargo tanks protected by a double hull which extends for the entire length of the cargo area, consisting of double sides and double bottom spaces for the carriage of water ballast or void spaces.
1.2.2 A ballast tank is a tank which is used solely for water ballast.
1.2.3 Overall survey is a survey intended to report on the overall condition of the hull structure and determine the extent of additional close-up surveys.
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(adopted on 20 May 2005)
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1.2.4 Close-up survey is a survey where the details of structural components are within

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1.2.4 Close-up survey is a survey where the details of structural components are within the close visual inspection range ofthe surveyor, i.e. preferably within reach of hand.

1.2.5 Transverse section includes all longitudinal members such as plating, longitudinals and girders at the deck, sides, bottom, inner bottom and longitudinal bulkheads.

1.2.6 Representative tanks are those which are expected to reflect the condition of other tanks of similar type and service and with similar corrosion protection systems. When selecting representative tanks account should be taken of the service and repair history on board and identifiable critical and/or suspect areas.

1.2.7 Suspect areas are locations showing Substantial Corrosion and/or are considered by the surveyor to be prone to rapid wastage.

1.2.8 Substantial corrosion is an extent of corrosion such that assessment of corrosion pattern indicates a wastage in excess of 75% of allowable margins, but within acceptable limits.

by the surveyor to be prone to rapid wastage.

1.2.8 Substantial corrosion is an extent of corrosion such that assessment of corrosion pattern indicates a wastage in excess of 75% of allowable margins, but within acceptable limits.

1.2.9 Corrosion prevention system is normally considered either:

.1 a full hard coating, or

.2 a full hard coating supplemented by anodes.

Protective coating should usually be epoxy coating or equivalent. Other coating systems may be considered acceptable as alternatives provided that they are applied and maintained in compliance with the manufacturer's specification.

Where soft coatings have been applied, safe access should be provided for the surveyor to verify the effectiveness of the coating and to carry out an assessment of the condition of internal structures which may include spot removal of the coating. When safe access cannot be provided, the soft coating should be removed.

1.2.10 Coating condition is defined as follows:

GOOD condition with only minor spot rusting,

FAIR condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of

POOR

1.2.10 Coating condition is defined as follows:
GOOD
FAIR
POOR
condition with only minor spot rusting,
condition with local breakdown of coating at edges of stiffeners
and weld connections and/or light rusting over 20% or more of
areas under consideration, but less than as defined for POOR
condition,
condition with general breakdown of coating over 20% or more of
areas or hard scale at 10% or more of areas under consideration.
1.2.11 Critical structural areas are locations which have been identified from
calculations to require monitoring or from the service history of the subject ship or from
similar or sister ships to be sensitive to cracking, buckling or corrosion which would
impair the structural integrity ofthe ship.
1.2.12 Cargo area is an area as defined in regulation II-2/3.6 of the 1974 SOLAS
Convention, as amended.
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1.2.13 Intermediate survey is a survey carried out either at the second or the third
annual surveyor between these surveys.

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1.2.13 Intermediate survey is a survey carried out either at the second or the third annual surveyor between these surveys.

1.2.14 A prompt and thorough repair is a permanent repair completed at the time of survey to the satisfaction of the Surveyor, therein removing the need for the imposition of any associated condition of classification or recommendation.

1.2.15 Specially considered means sufficient close-up inspection and thickness measurements are taken to confirm the actual average condition of the structure under coating.

1.3 Repairs

1.3.1 Any damage in association with wastage over the allowable limits (including buckling, grooving, detachment or fracture), or extensive areas of wastage over the allowable limits, which affects or, in the opinion of the Administration, will affect the ship's structural, watertight or weathertight integrity, should be promptly and thoroughly (see 1.2.14) repaired. Areas to be considered include:

.1 bottom structure and bottom plating;

.2 side structure and side plating;

.3 deck structure and deck plating;

(see 1.2.14) repaired. Areas to be considered include:
.1 bottom structure and bottom plating;
.2 side structure and side plating;
.3 deck structure and deck plating;
.4 inner bottom structure and inner bottom plating;
.5 inner side structure and inner side plating;
.6 longitudinal buJkhead(s) structure and longitudinal bulkhead(s) plating, where fitted;
.7 transverse watertight or oiltight bulkheads structure and transverse watertight or oiltight bulkheads plating;
.8 hatch covers or hatch coamings, where fitted; and
.9 items in paragraph 3.3.
Where adequate repair facilities are not available, the Administration may allow the ship to proceed directly to a repair facility. This may require discharging the cargo and/or temporary repairs for the intended voyage.
1.3.2 Additionally, when a survey results in the identification of corrosion or structural defects, either of which, in the opinion of the Administration, will impair the ship's fitness for continued service, remedial measures should be implemented before the ship continues in service.
1.4 Surveyors
For tankers of 20,000 tons deadweight and above, two surveyors should jointly carry out

fitness for continued service, remedial measures should be implemented before the ship continues in service.

1.4 Surveyors

For tankers of 20,000 tons deadweight and above, two surveyors should jointly carry out the first scheduled renewal survey after the tanker passes 10 years of age, and all subsequent renewal surveys and intermediate surveys. If the surveys are carried out by a

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recognized organization, the surveyors should be exclusively employed by such recognized organizations.

2 Renewal survey

2.1 General

2.1.1 The renewal survey may be commenced at the fourth annual survey and be progressed during the succeeding year with a view to completion by the fifth anniversary date.

2.1.2 As part of the preparation for the renewal survey the survey programme should be dealt with in advance of the renewal survey. The thickness measurement should not be carried out before the fourth annual survey.

fifth anniversary date.

2.1.2 As part of the preparation for the renewal survey the survey programme should be dealt with in advance of the renewal survey. The thickness measurement should not be carried out before the fourth annual survey.

2.1.3 The survey should include, in addition to the requirements of the annual survey, examination, tests and checks of sufficient extent to ensure that the hull and related piping as required in 2.1.5 is in a satisfactory condition and is fit for its intended purpose for the new period of validity of the Cargo Ship Safety Construction Certificate, subject to proper maintenance and operation and to periodical surveys being carried out.

2.1.4 All cargo tanks, ballast tanks, pump-rooms, pipe tunnels, cofferdams and void spaces bounding cargo tanks, decks and outer hull should be examined, and this examination should be supplemented by thickness measurement and testing as deemed necessary, to ensure that the structural integrity remains effective. The examination should be sufficient to discover substantial corrosion, significant deformation, fractures, damages or other structural deterioration.

necessary, to ensure that the structural integrity remains effective. The examination should be sufficient to discover substantial corrosion, significant deformation, fractures, damages or other structural deterioration.

2.1.5 Cargo piping on deck, including crude oil washing (COW) piping, and cargo and ballast piping within the above tanks and spaces should be examined and operationally tested to working pressure to attending surveyor's satisfaction to ensure that tightness and condition remain satisfactory. Special attention should be given to any ballast piping in cargo tanks and cargo piping in ballast tanks and void spaces, and surveyors should be advised on all occasions when this piping, including valves and fittings, are open during repair periods and can be examined internally.

2.2 Dry-dock survey

2.2.1 A survey in dry-dock should be a part of the renewal survey. There should be a minimum of two inspections of the outside of the ship's bottom during the five-year period ofthe Safety Construction Certificate. In all cases, the maximum interval between bottom inspections should not exceed 36 months.

minimum of two inspections of the outside of the ship's bottom during the five-year period ofthe Safety Construction Certificate. In all cases, the maximum interval between bottom inspections should not exceed 36 months.

2.2.2 For ships of 15 years of age and over, inspection of the outside of the ship's bottom should be carried out with the ship in dry-dock. For ships ofless than 15 years of age, alternate inspections of the ship's bottom not conducted in conjunction with the renewal survey may be carried out with the ship afloat. Inspection of the ship afloat should only be carried out when the conditions are satisfactory and the proper equipment and suitably qualified staff is available.

2.2.3 If a survey in dry-dock is not completed in conjunction with the renewal surveyor if the 36 month maximum interval referred to in 2.2.1 is not complied with, the Cargo Ship Safety Construction Certificate should cease to be valid until a survey in dry-dock is completed.

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completed.
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2.3 Tank corrosion prevention system
Where provided, the condition of the corrosion prevention system of cargo tanks should be examined. A ballast tank where a protective coating is found in POOR condition and it is not renewed, or where soft coating has been applied, or where a protective coating has not been applied from the time of construction, the tank in question should be examined at annual intervals. Thickness measurement should be carried out as deemed necessary by the surveyor.
2.4 Extent of overall and close-up surveys
2.4.1 An overall survey of all integral tanks and spaces should be carried out at the renewal survey.
2.4.2 The requirements for close-up surveys at the renewal survey are given in annex 1.
2.4.3 The surveyor may extend the scope of the close-up survey as deemed necessary taking into account the maintenance of the tanks under survey, the condition of the corrosion prevention system and also in the following cases:

2.4.3 The surveyor may extend the scope of the close-up survey as deemed necessary taking into account the maintenance of the tanks under survey, the condition of the corrosion prevention system and also in the following cases:
.1 in particular, tanks having structural arrangements or details which have suffered defects in similar tanks or on similar ships according to available information;
.2 in tanks which have structures with reduced scantlings in association with a corrosion prevention system approved by the Administration.
2.4.4 For areas in tanks where coatings are found to be in GOOD condition as defined in 1.2.10, the extent of close-up surveys according to annex 1 may be specially considered by the Administration.
2.5 Extent of thickness measurements
2.5.1 The requirements for thickness measurements at the renewal survey are given in annex 2.
2.5.2 Where substantial corrosion as defined in 1.2.8 is found, the extent of thickness measurements should be increased in accordance with the requirements of annex 4.
2.5.3 The surveyor may extend the thickness measurements as deemed necessary.

2.5.2 Where substantial corrosion as defined in 1.2.8 is found, the extent of thickness measurements should be increased in accordance with the requirements of annex 4.
2.5.3 The surveyor may extend the thickness measurements as deemed necessary.
2.5.4 For areas in tanks where coatings are found to be in GOOD condition as defined in 1.2.10, the extent of thickness measurements according to annex 2 may be specially considered by the Administration.
2.5.5 Transverse sections should be chosen where the largest reductions are suspected to occur or are revealed from deck plating measurements.
2.5.6 In cases where two or three sections are to be measured, at least one should include a ballast tank within O.SL amidships.
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2.6 Extent of tank pressure testing
2.6.1 The requirements for tank pressure testing at the renewal survey are given in annex 3.
2.6.2 The surveyor may extend the tank pressure testing as deemed necessary.

2.6 Extent of tank pressure testing
2.6.1 The requirements for tank pressure testing at the renewal survey are given in annex 3.
2.6.2 The surveyor may extend the tank pressure testing as deemed necessary.
2.6.3 Generally, the pressure should correspond to a water level to the top of access hatches for cargo tanks, or top of air pipes for ballast tanks.
3 Annual survey
3.1 General
The annual survey should consist of an examination for the purpose of ensuring, as far as practicable, that the hull and piping are maintained in a satisfactory condition and should take into account the service history, condition and extent of the corrosion prevention system of ballast tanks and areas identified in the survey report file.
3.2 Examination of the hull
3.2.1 Examination of the hull plating and its closing appliances should be carried out as far as can be seen.
3.2.2 Examination of watertight penetrations should be carried out as far as practicable.
3.3 Examination of weather decks
3.3.1 Examination of cargo tank openings including gaskets, covers, coammgs and flame screens.
3.3.2 Examination of cargo tank pressure/vacuum valves and flame screens.

3.3 Examination of weather decks
3.3.1 Examination of cargo tank openings including gaskets, covers, coammgs and flame screens.
3.3.2 Examination of cargo tank pressure/vacuum valves and flame screens.
3.3.3 Examination of flame screens on vents to all bunker and oily slop tanks.
3.3.4 Examination of cargo, crude oil washing, bunker and vent piping systems, including vent masts and headers.
3.4 Examination of cargo pump-rooms and pipe tunnels
3.4.1 Examination of all bulkheads for signs of oil leakage or fractures and, m particular, the sealing arrangements of all penetrations of bulkheads.
3.4.2 Examination of the condition of all piping systems and pipe tunnels.
3.5 Examination of ballast tanks
3.5.1 Examination of ballast tanks should be carried out when required as a consequence of the results of the renewal survey and intermediate survey. When extensive corrosion is found, thickness measurements should be carried out.
3.5.2 Where substantial corrosion as defined in 1.2.8 is found, the extent of thickness measurements should be increased in accordance with the requirements in annex 4.
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3.5.2 Where substantial corrosion as defined in 1.2.8 is found, the extent of thickness measurements should be increased in accordance with the requirements in annex 4.
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3.5.3 Double hull oil tankers exceeding 15 years of age
3.5.3.1 All ballast tanks adjacent to (i.e. with a common plane boundary) a cargo or fuel tank with any means of heating should be examined internally. When considered necessary by the surveyor, thickness measurement should be carried out and if the results of these thickness measurements indicate that substantial corrosion is found, the extent of thickness measurements should be increased in accordance with annex 4.
3.5.3.2 Ballast tanks which were found, at the previous intermediate or renewal survey, to have no substantial corrosion within the tank and which were found in compliance with either of the following conditions:
.1 coating in GOOD condition; or
.2 coating of the common boundary, including adjacent structures, in GOOD

to have no substantial corrosion within the tank and which were found in compliance with either of the following conditions:

.1 coating in GOOD condition; or

.2 coating of the common boundary, including adjacent structures, in GOOD condition and the coating of the remaining parts of the tank in FAIR condition,

may be specially considered by the Administration.

4 Intermediate survey

4.1 General

4.1.1 Items that are additional to the requirements of the annual survey may be surveyed either at the second or third annual surveyor between these surveys.

4.1.2 The survey extent of cargo and ballast tanks dependent on the age of the ship is specified in 4.2, 4.3 and 4.4 and shown in annex 5.

4.1.3 For weather decks, an examination as far as applicable of cargo, crude oil washing, bunker, ballast, steam and vent piping systems as well as vent masts and headers. If upon examination there is any doubt as to the condition of the piping, the piping may be required to be pressure tested, thickness measured or both.

4.2 Oil tankers of 5 to 10 years of age

4.2.1 The requirements of 4.1.3 apply.

headers. If upon examination there is any doubt as to the condition of the piping, the piping may be required to be pressure tested, thickness measured or both.

4.2 Oil tankers of 5 to 10 years of age

4.2.1 The requirements of 4.1.3 apply.

4.2.2 For tanks used for salt water ballast, an overall survey of representative tanks selected by the surveyor should be carried out. If the overall survey of salt water ballast tanks reveals no visible structural defects, the examination may be limited to verification that the protective coatings remain efficient.

4.2.3 Where POOR coating condition, corrosion or other defects are found in salt water ballast tanks or where a protective coating was not applied from the time of construction, the examination should be extended to other ballast tanks of the same type.

4.2.4 In salt water ballast tanks where a protective coating is found in POOR condition and it is not renewed, where soft coating has been applied, or where a protective coating was not applied from the time of construction, the tanks in question should be examined and thickness measurements carried out as considered necessary at annual intervals.

was not applied from the time of construction, the tanks in question should be examined and thickness measurements carried out as considered necessary at annual intervals.

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4.3 Oil tankers of 10 to 15 years of age

4.3.1 The requirements of 4.2 apply.

4.3.2 An overall survey of at least two representative cargo tanks should be carried out.

4.3.3 For ballast tanks, where fitted, an overall survey of all such tanks should be carried out. If such survey reveals no visible structural defects, the survey may be limited to a verification that the protective coatings remain efficient.

4.3.4 Extent of close-up survey

Ballast tanks: to the same extent as previous renewal survey;

Cargo tanks: the extent of survey should be based on the record of the previous renewal survey and repair history of the tanks, and be applied to two cargo tanks after the second renewal survey.

The minimum requirements of close-up surveys are given in annex 5. The extent of

survey and repair history of the tanks, and be applied to two cargo tanks after the second renewal survey.

The minimum requirements of close-up surveys are given in annex 5. The extent of close-up surveys may be extended as stated in 2.4.3. For areas in tanks where coatings are found to be in GOOD condition, the extent of the close-up surveys according to annex 5 may be specially considered by the Administration.

4.3.5 Extent of thickness measurements

The extent of thickness measurements is also given in annex 5. The mllllmum requirements for thickness measurements at the intermediate survey are areas found to be suspect areas according to 1.2.7 at the previous renewal survey. Where substantial corrosion as defined in 1.2.8 is found, the extent of the thickness measurements according to annex 5 should be increased in accordance with the requirements of annex 4.

4.4 Oil tankers exceeding 15 years of age

The requirements of the intermediate survey should be to the same extent as the previous renewal survey as required in 2 and 5.1. However, pressure testing of cargo and ballast tanks is not required unless deemed necessary by the attending surveyor.

renewal survey as required in 2 and 5.1. However, pressure testing of cargo and ballast tanks is not required unless deemed necessary by the attending surveyor.

5 Preparations for survey

5.1 Survey programme

5.1.1 A specific survey programme should be worked out in advance of the renewal survey by the owner in co-operation with the Administration. The survey programme should be in a written format based on the information in annex 6A. The survey should not commence until the survey programme has been agreed.

5.1.1.1 Prior to the development of the survey progra=e , the survey planning questionnaire should be completed by the owner based on the information set out in annex 6B, and forwarded to the Administration.

5.1.2 In developing the survey programme, the following documentation should be collected and consulted with a view to selecting tanks, areas, and structural elements to be examined:

.1 survey status and basic ship information;

5.1.2 In developing the survey programme, the following documentation should be collected and consulted with a view to selecting tanks, areas, and structural elements to be examined:
.1 survey status and basic ship information;
.2 documentation on board, as described in 6.2 and 6.3;
.3 main structural plans of cargo and ballast tanks (scantlings drawings), including information regarding use of high-tensile steels (HTS);
.4 Condition Evaluation Report, according to annex 9;
.5 relevant previous damage and repair history;
.6 relevant previous survey and inspection reports from both the recognized organization and the owner;
.7 cargo and ballast history for the last 3 years, including carriage of cargo under heated conditions;
.8 details of the inert gas plant and tank cleaning procedures;
.9 information and other relevant data regarding conversion or modification of the ship's cargo and ballast tanks since the time of construction;
.10 description and history of the coating and corrosion protection system (including anodes and previous class notations), if any;
.11 inspections of the Owner's personnel during the last 3 years with reference

.10 description and history of the coating and corrosion protection system (including anodes and previous class notations), if any;
.11 inspections of the Owner's personnel during the last 3 years with reference to structural deterioration in general, leakages in tank boundaries and piping and condition of the coating and corrosion protection system (including anodes), if any. A guidance for reporting is shown in annex 6C;
.12 information regarding the relevant maintenance level during operation including port state control reports of inspection containing hull-related deficiencies , Safety Management System non-conformities relating to hull maintenance, including the associated corrective action(s); and
.13 any other information that will help identify suspect areas and critical structural areas.

5.1.3 The submitted survey programme should account for and comply, as a minimum, with the requirements of annexes 1, 2 and 3 and paragraph 2.6 for close-up survey, thickness measurement and tank testing, respectively, and should include relevant information including at least:
.1 basic ship information and particulars;

with the requirements of annexes 1, 2 and 3 and paragraph 2.6 for close-up survey, thickness measurement and tank testing, respectively, and should include relevant information including at least:

.1 basic ship information and particulars;

.2 main structural plans of cargo and ballast tanks (scantling drawings), including information regarding use of high tensile steels (HTS);

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.3 arrangement of tanks ;

.4 list of tanks with information on their use, extent of coatings and corrosion protection systems;

.5 conditions for survey (e.g., information regarding tank cleaning , gas freeing, ventilation, lighting, etc.);

.6 provisions and methods for access to structures;

.7 equipment for surveys;

.8 identification of tanks and areas for close-up survey (see 2.4) ;

.9 identification of areas and sections for thickness measurement (see 2.5);

.10 identification of tanks for tank testing (see 2.6);

.11 identification of the thickness measurement company;

.9 identification of areas and sections for thickness measurement (see 2.5);
.10 identification of tanks for tank testing (see 2.6);
.11 identification of the thickness measurement company;
.12 damage experience related to the ship in question; and
.13 critical structural areas and suspect areas, where relevant.

5.1.4 The Administration will advise the owner of the maximum acceptable structural corrosion diminution levels applicable to the ship.

5.1.5 Use may also be made of the Guidelines for technical assessment in conjunction with the planning of enhanced surveys for tankers, contained in annex II. These Guidelines are a recommended tool which may be invoked at the discretion of the Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

5.2 Conditions for survey

5.2.1 The owner should provide the necessary facilities for a safe execution of the survey.

5.2.1.1 In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access should be agreed between the owner and the Administration.

survey.
5.2.1.1 In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access should be agreed between the owner and the Administration.
5.2.1.2 Details of the means of access should be provided in the survey planning questionnaire.
5.2.1.3 In cases where the provisions of safety and required access are judged by the attending surveyors not to be adequate, the survey of the spaces involved should not proceed.
5.2.2 Tanks and spaces should be safe for access. Tanks and spaces should be gas free and properly ventilated. Prior to entering a tank, void or enclosed space, it should be verified that the atmosphere in that space is free from hazardous gas and contains sufficient oxygen .
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5.2.3 Tanks and spaces should be sufficiently clean and free from water, scale, dirt, oil residues, sediments, etc., to reveal corrosion, deformation, fractures, damages or other

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5.2.3 Tanks and spaces should be sufficiently clean and free from water, scale, dirt, oil residues, sediments, etc., to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating. In particular this applies to areas which are subject to thickness measurement.

5.2.4 Sufficient illumination should be provided to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition ofthe coating.

5.2.5 The surveyor(s) should always be accompanied by at least one responsible person, assigned by the owner, experienced in tank and enclosed spaces inspection. In addition a backup team of at least two experienced persons should be stationed at the hatch opening of the tank or space that is being surveyed. The back-up team should continuously observe the work in the tank or space and should keep lifesaving and evacuation equipment ready for use.

5.2.6 A communication system should be arranged between the survey party in the tank or space being examined, the responsible officer on deck and, as the case may be, the

equipment ready for use.
5.2.6 A communication system should be arranged between the survey party in the tank or space being examined, the responsible officer on deck and, as the case may be, the navigation bridge. The communication arrangements should be maintained throughout the survey.
5.3 Access to structures
5.3.1 For overall surveys, means should be provided to enable the surveyor to examine the structure in a safe and practical way.
5.3.2 For close-up surveys, one or more of the following means for access, acceptable to the surveyor, should be provided:
.1 permanent staging and passages through structures;
.2 temporary staging and passages through structures;
.3 lifts and moveable platforms;
.4 boats or rafts;
.5 portable ladders;
.6 other equivalent means.
5.4 Equipment for survey
5.4.1 Thickness measurement should normally be carried out by means of ultrasonic test equipment. The accuracy of the equipment should be proven to the surveyor as required.
5.4.2 One or more of the following fracture detection procedures may be required if deemed necessary by the surveyor:
.1 radiographic equipment;
.2 ultrasonic equipment;

required.
5.4.2 One or more of the following fracture detection procedures may be required if deemed necessary by the surveyor:
.1 radiographic equipment;
.2 ultrasonic equipment;
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.3 magnetic particle equipment;
.4 dye penetrant; and
.5 other equivalent means.
5.4.3 Explosimeter, oxygen-meter, breathing apparatus, lifelines, riding belts with rope and hook and whistles together with instructions and guidance on their use should be made available during the survey. A safety checklist should be provided.
5.4.4 Adequate and safe lighting should be provided for the safe and efficient conduct of the survey.
5.4.5 Adequate protective clothing should be made available and used during the survey (e.g., safety helmet, gloves, safety shoes, etc.).
5.5 Surveys at sea or at anchorage
5.5.1 Surveys at sea or at anchorage may be accepted provided the surveyor(s) is given the necessary assistance from the personnel on board. Necessary precautions and

5.5 Surveys at sea or at anchorage
5.5.1 Surveys at sea or at anchorage may be accepted provided the surveyor(s) is given the necessary assistance from the personnel on board. Necessary precautions and procedures for carrying out the survey should be in accordance with 5.1,5.2, 5.3 and 5.4.
5.5.2 A communication system should be arranged between the survey party in the tank and the responsible officer on deck. This system should also include the personnel in charge of ballast pump handling if boats or rafts are used.
5.5.3 Surveys of tanks by means of boats or rafts may only be undertaken with the agreement of the surveyor, who should take into account the safety arrangements provided, including weather forecasting and ship response in reasonable sea conditions.
5.5.4 When rafts or boats are used for close-up surveys, the following conditions should be observed:
.1 only rough duty, inflatable rafts or boats, having satisfactory residual buoyancy and stability even if one chamber is ruptured, should be used;
.2 the boat or raft should be tethered to the access ladder and an additional person should be stationed down the access ladder with a clear view of the

buoyancy and stability even if one chamber is ruptured, should be used;
.2 the boat or raft should be tethered to the access ladder and an additional person should be stationed down the access ladder with a clear view of the boat or raft;
.3 appropriate lifejackets should be available for all participants;
.4 the surface of water in the tank should be calm (under all foreseeable conditions the expected rise of water within the tank should not exceed 0.25 m) and the water level either stationary or falling. On no account should the level of the water be rising while the boat or raft is in use;
.5 the tank or space must contain clean ballast water only. Even a thin sheen of oil on the water is not acceptable;
.6 at no time should the water level be allowed to be within I m of the deepest under-deck web face flat so that the survey team is not isolated
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from a direct escape route to the tank hatch. Filling to levels above the

from a direct escape route to the tank hatch. Filling to levels above the deck transverses should only be contemplated if a deck access manhole is fitted and open in the bay being examined, so that an escape route for the survey party is available at all times. Other effective means of escape to the deck may be considered;

.7 if the tanks (or spaces) are connected by a common venting system, or inert gas system, the tank in which the boat or raft should be used should be isolated to prevent a transfer of gas from other tanks (or spaces).

5.5.5 Rafts or boats alone may be allowed for inspection of the under-deck areas of tanks or spaces if the depth of the webs is 1.5 m or less.

5.5.6 If the depth of the webs is more than 1.5 m, rafts or boats alone may be allowed only:

. I when the coating of the under-deck structure is in GOOD condition and there is no evidence of wastage; or

.2 if a permanent means of access is provided in each bay to allow safe entry

only:
. I when the coating of the under-deck structure is in GOOD condition and there is no evidence of wastage; or
.2 if a permanent means of access is provided in each bay to allow safe entry and exit. This means of access should be direct from the deck via a vertical ladder with a small platform fitted approximately 2 m below the deck. Other effective means of escape to the deck may be considered.
If neither of the above conditions are met, then staging or other equivalent means should be provided for the survey of the under-deck areas.
5.5.7 The use of rafts or boats alone in 5.5.5 and 5.5.6 does not preclude the use of boats or rafts to move about within a tank during a survey.
5.6 Survey planning meeting
5.6. I Proper preparation and close co-operation between the attending surveyor(s) and the owner's representatives on board prior to and during the survey are an essential part in the safe and efficient conduct of the survey. During the survey on board safety meetings should be held regularly.
5.6.2 Prior to commencement of any part of the renewal and intermediate survey, a

the safe and efficient conduct of the survey. During the survey on board safety meetings should be held regularly.

5.6.2 Prior to commencement of any part of the renewal and intermediate survey, a survey planning meeting should be held between the attending surveyor(s), the owner's representative in attendance, the thickness measurement company operator (as applicable) and the master of the ship for the purpose of ascertaining that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

5.6.3 The following is an indicative list of items that should be addressed III the meeting:

.I schedule of the vessel (i.e. the voyage, docking and undocking manoeuvres, periods alongside, cargo and ballast operations, etc.);

.2 provisions and arrangements for thickness measurements (i.e. access, cleaning/de-scaling, illumination, ventilation, personal safety);

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.3 extent of the thickness measurements;
.4 acceptance criteria (refer to the list of minimum thicknesses);
.5 extent of close-up survey and thickness measurement considering the coating condition and suspect areas/areas of substantial corrosion;
.6 execution of thickness measurements;
.7 taking representative readings III general and where uneven corrosion/pitting is found;
.8 mapping of areas of substantial corrosion; and
.9 communication between attending surveyor(s) the thickness measurement company operator(s) and owner representative(s) concerning findings.
6 Documentation on board
6.1 General
6.1.1 The owner should obtain, supply and maintain on board the ship documentation as specified in 6.2 and 6.3, which should be readily available for the surveyor. The condition evaluation report referred to in 6.2 should include a translation into English.
6.1.2 The documentation should be kept on board for the lifetime of the ship.
6.2 Survey report file

condition evaluation report referred to in 6.2 should include a translation into English.
6.1.2 The documentation should be kept on board for the lifetime of the ship.
6.2 Survey report file
6.2.1 A survey report file should be a part of the documentation on board consisting of:
.1 reports of structural surveys (annex 8);
.2 condition evaluation report (annex 9); and
.3 thickness measurement reports (annex 10).
6.2.2 The survey report file should be available also III the owner's and the Administration's offices.
6.3 Supporting documents
The following additional documentation should be available on board:
.1 all documents required by 5.1.2;
.2 survey programme as required by 5.1 until such time as the renewal survey has been completed; and
.3 any other information that would help to identifY critical structural areas and/or suspect areas requiring inspection.
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6.4 Review of documentation on board

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6.4 Review of documentation on board
Prior to survey, the surveyor should examine the completeness of the documentation on board and its contents as a basis for the survey.
7 Procedures for thickness measurements
7.1 General
7.1.1 The required thickness m easurements, if not carried out by the recognized organization acting on behalf of the Administration, should be witnessed by a surveyor of the recognized organization. The surveyor should be on board to the extent necessary to control the process.
7.1.2 The thickness measurement company should be part of the survey planning meeting to be held prior to commencing the survey.
7. 1.3 In all cases the extent of the thickness measurements should be sufficient as to represent the actual average condition.
7.2 Certification of thickness measurement company
The thickness measurements should be carried out by a qualified company certified by an

represent the actual average condition.

7.2 Certification of thickness measurement company

The thickness measurements should be carried out by a qualified company certified by an organization recognized by the Administration according to principles stated in annex 7.

7.3 Reporting

7.3.1 A thickness measurement report should be prepared and submitted to the Administration. The report should give the location of measurements, the thickness measured as well as corresponding original thickness. Furthermore, the report should give the date when the measurements were carried out, type of measuring equipment, names of personnel and their qualifications and be signed by the operator. The thickness measurement report should follow the principles as specified in the recommended procedures for thickness measurements set out in annex 10.

7.3.2 The surveyor should verify and countersign the thickness measurement reports.

8 Reporting and evaluation of survey

8.1 Evaluation of survey report

8.1.1 The data and information on the structural condition of the ship collected during the survey should be evaluated for acceptability and continued structural integrity of the ship.

8.1 Evaluation of survey report
8.1.1 The data and information on the structural condition of the ship collected during the survey should be evaluated for acceptability and continued structural integrity of the ship.
8.1.2 In case of oil tankers of 130 m in length and upwards (as defined in the International Convention on Load Lines in force), the ship's longitudinal strength should be evaluated by using the thickness of structural members measured, renewed and reinforced, as appropriate, during the renewal survey of safety construction carried out after the ship reached 10 years of age, in accordance with the criteria for longitudinal strength ofthe ship's hull girder for oil tankers specified in annex 12.
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8.1.3 The analysis of data should be carried out and endorsed by the Administration and the conclusions of the analysis should form a part of the condition evaluation report.

8.1.3 The analysis of data should be carried out and endorsed by the Administration and the conclusions of the analysis should form a part of the condition evaluation report.
8.1.4 The final result of the evaluation of the ship's longitudinal strength required in 8.1.2, after renewal or reinforcement work of structural members, if carried out as a result of initial evaluation, should be reported as a part of the condition evaluation report.
8.2 Reporting
8.2.1 Principles for survey reporting are shown in annex 8.
8.2.2 When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items examined and/or tested (pressure testing, thickness measurements, etc.) and an indication of whether the item has been credited, should be made available to the next attending surveyor(s), prior to continuing or completing the survey.
8.2.3 A condition evaluation report of the survey and results should be issued to the owner as shown in annex 9 and placed on board the ship for reference at future surveys.

completing the survey.
8.2.3 A condition evaluation report of the survey and results should be issued to the owner as shown in annex 9 and placed on board the ship for reference at future surveys. The condition evaluation report should be endorsed by the Administration.
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ANNEX I
MINIMUM REQUIREMENTS FOR CLOSE -UP SURVEY AT RENEWAL SURVEY OF DOUBLE HULL OIL TANKERS
Age < 5 years 5<age < 10 years IO<age<15 years Age> 15 years
I 2 3 4
One web frame (I) , All web frames (I) , in All web frames (1), As for ships in a complete ballast a complet e ballast in all ballast tanks referred to in tank tank column 3
(see Note 1) (see Note I)
The knuckle area and the upper part (5 m approximately) of one transverse web frame in each remaining ballast tank Additional areas as deemed necessary by the Administration
(6)
One deck transverse, One deck transverse, All web frames (7), in a cargo oil tank (2) in two cargo oil tanks including deck (2) transverse and cross

web frame in each as deemed
remaining ballast tank necessary by the
(6) Administration
One deck transverse, One deck transverse, All web frames (7),
in a cargo oil tank (2) in two cargo oil tanks including deck
(2) transverse and cross
ties, if fitted , in a
cargo oil tank
One web frame (7),
including deck
transverse and cross
ties, if fitted, in each
remaining cargo oil
tank
One transverse One transverse All transverse
bulkhead (4), in a bulkhead (4), in each bulkhead s, in all
complete ballast tank complete ballast tank cargo oil (3) and
(see Note 1) (see Note 1) ballast (4) tanks
One transverse One transverse
bulkhead (5) in a bulkhead (5), in two
cargo oil centre tank cargo oil centre tanks
One transverse One transverse
bulkhead (5), in a bulkhead (5), in a
cargo oil wing tank cargo oil wing tank
(see Note 2) (see Note 2)
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NOTES:
(1), (2), (3), (4), (5), (6) and (7) are areas to be subjected to close-up surveys and thickness

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NOTES:
(1), (2), (3), (4), (5), (6) and (7) are areas to be subjected to close-up surveys and thickness measurements (see appendix 3 of annex 10).
(1) Web frame in a ballast tank means vertical web in side tank, hopper web in hopper tank, floor in double bottom tank and deck transverse in double deck tank (where fitted), including adjacent sttuctural members. In fore and aft peak tanks web frame means a complete transverse web frame ring including adjacent sttuctural members.
(2) Deck transverse, including adjacent deck structural members (or external sttucture on deck in way of the tank, where applicable).
(3) Transverse bulkhead complete in cargo tanks, including girder system, adjacent structural members (such as longitudinal bulkheads) and internal sttucture of lower and upper stools, where fitted.
(4) Transverse bulkhead complete in ballast tanks, including girder system and adjacent sttuctural members, such as longitudinal bulkheads, girders in double bottom tanks, inner bottom plating, hopper side, connecting brackets.

(4) Transverse bulkhead complete in ballast tanks, including girder system and adjacent sttuctural members, such as longitudinal bulkheads, girders in double bottom tanks, inner bottom plating, hopper side, connecting brackets.
(5) Transverse bulkhead lower part in cargo tank, including girder system, adjacent sttuctural members (such as longitudinal bulkheads) and internal sttucture of lower stool, where fitted.
(6) The knuckle area and the upper part (5 metres approximately), including adjacent structural members. Knuckle area is the area of the web frame around the connections of the slope hopper plating to the inner hull bulkhead and the inner bottom plating, up to 2 metres from the corners both on the bulkhead and the double bottom.
(7) Web frame in a cargo oil tank means deck transverse, longitudinal bulkhead vertical girder and cross ties, where fitted, including adjacent structural members.
Note 1 Complete ballast tank: means double bottom tank plus double side tank plus double deck tank, as applicable, even if these tanks are separate.
Note 2 Where no centre cargo tanks are fitted (as in the case of centre longitudinal

Note 1 Complete ballast tank: means double bottom tank plus double side tank plus double deck tank, as applicable, even if these tanks are separate.
Note 2 Where no centre cargo tanks are fitted (as in the case of centre longitudinal bulkhead), transverse bulkheads in wing tanks should be surveyed.
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ANNEX 2
MINIMUM REQUIREMENTS FOR THICKNESS MEASUREMENTS AT RENEWAL SURVEY OF DOUBLE HULL OIL TANKERS

Age<S years	S<age< I 0 years	IO<age< IS years	Age> 15 years
I	2	3	4
One section of deck plating for the full beam of the ship within the cargo area	Within the cargo area: each deck plate one transverse section (1)	Within the cargo area: each deck plate two transverse sections (1)	Within the cargo area: each deck plate three transverse sections (1)
all wind and water strakes outside the cargo	each bottom plate all wind and water strakes		
Selected wind and water strakes outside the cargo	Selected wind and water strakes outside the cargo	Selected wind and water strakes outside the cargo	

all wind and each bottom plate
water strakes all wind and
water strakes
Selected wind and Selected wind and Selected wind and
water strakes water strakes water strakes
outside the cargo outside the cargo outside the cargo
area area area
~easurements, for ~easurements, for ~easurements, for ~easurements , for
general assessment general assessment general assessment general assessment
and recording of and recording of and recording of and recording of
corrosion pattern, of corrosion pattern, of corrosion pattern, of corrosion pattern, of
those structura l those structural those structural those structural
members subject to members subject to members subject to members subject to
close-up survey close-up survey close-up survey close-up survey
according to according to according to according to
annex 1 annex I annex I annex I
Suspect areas Suspect areas Suspect areas Suspect areas
(1): at least one section should be within O.SL amidships.
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ANNEX 3

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ANNEX 3
MINIMUM REQUIREMENTS FOR TANK TESTING AT RENEWAL SURVEY
OF DOUBLE HULL OIL TANKERS
Age,::::5 years 5<age.::::10 years Age> I 0 years
I 2 3
All ballast tank boundaries All ballast tank boundarie s All ballast tank boundaries
Cargo tank boundaries Cargo tank boundarie s Cargo tank boundaries
facing ballast tanks, void facing ballast tanks, void facing ballast tanks, void
spaces, pipe tunnels, spaces, pipe tunnels, spaces, pipe tunnels,
representative fuel oil representative fuel oil representative fuel oil
tanks , pump-rooms or tanks, pump-room s or tanks, pump-rooms or
cofferdams cofferdams cofferdams
All cargo tank bulkhead s All remaining cargo tank
which form the boundaries bulkhead s
of segregated cargoes
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ANNEX 4/Sheet 1

of segregated cargoes
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ANNEX 4/Sheet 1
REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT
AREAS OF SUBSTANTIAL CORROSION
RENEWAL SURVEY OF DOUBLE HULL OIL TANKERS
BOTTOM, INNER BOTTOM AND HOPPER STRUCTURE

Structural member	Extent of measurement	Pattern of measurement
Bottom, inner bottom and hopper structure plating	Minimum of three bays across double bottom tank, including between longitudinals and aft bay	5-point pattern for each panel Measurements around floors and under all suction bell mouths
Bottom, inner bottom and hopper structure longitudinals where bottom plating measured	Minimum of three longitudinals in each bay	Three measurements in line across flange and three measurements on vertical web
Bottom girders, including the watertight ones	At fore and aft watertight floors and in centre of tanks	Vertical line of single measurement s on girder plating with one measurement between each panel stiffener, or a minimum of three measurements

watertight ones floors and in centre of tanks measurement s on girder
plating with one measurement
between each panel stiffener,
or a minimum of three
measurements
Bottom floors, including the Three floors in bays where 5-point pattern over two
watertight ones bottom plating measured , with square metre area
measurements at both ends
and middle
Hopper structure web frame Three floors in bays where 5-point pattern over one
nng bottom plating measured square metre of plating.
Single measurements on
flange
Hopper structure transverse - lower 1/3 of bulkhead 5-point pattern over one
watertight bulkhead or swash square metre of plating
bulkhead
- upper 2/3 of bulkhead 5-point pattern over two
square metre of plating
- stiffeners (minimum of For web, 5-point pattern over
three) span (two measurements
across web at each end and
one at centre of span). For
flange, single measurements
at each end and centre of span
Panel stiffening Where applicable Single measurements
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at each end and centre of span
Panel stiffening Where applicable Single measurements
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AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 49-
ANNEX 4/Sheet 2
REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT
AREAS OF SUBSTANTIAL CORROSION RENEWAL SURVEY OF DOUBLE
HULL OIL TANKERS WITHIN THE CARGO AREA LENGTH
DECK STRUCTURE
Structural member Extent of measurement Pattern of measurement
Deck plating Two transverse bands across Minimum of three
tank measurements per plate per
band
Deck longitudinals Every third longitudinal in Three measurements in line
each of two bands with a vertically on webs and two
minimum of one measurem ents on flange (if
longitudinal fitted)
Deck girders and brackets At fore and aft transverse Vertical line of single
(usually in cargo tanks only) bulkhead, bracket toes and in measurements on web plating
centre of tanks with one measurement
between each panel stiffener,
or a minimum of three
measurements.
Two measurements across
flange . 5-point pattern on
girderfbulkhead brackets

centre of tanks with one measurement
between each panel stiffener,
or a minimum of three
measurements.
Two measurements across
flange . 5-point pattern on
girderfbulkhead brackets
Deck transverse webs Minimum of two webs, with 5-point pattern over one
measurements at both ends square metre area.
and middle of span Single measurements on
flange
Vertical web and transverse Minimum of two webs, and 5-point pattern over one
bulkhead in wing ballast tank both transverse bulkheads square metre area
(two metres from deck)
Panel stiffening Where applicable Single measurements
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ANNEX 4/Sheet 3
REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT
AREAS OF SUBSTANTIAL CORROSION RENEWAL SURVEY OF DOUBLE
HULL OIL TANKERS WITHIN THE CARGO AREA LENGTH
STRUCTURE IN WING BALLAST TANKS
Structural member Extent of measurement Pattern of measurement
Side shell and longitudinal Single measurement
bulkhead plating:
- upper strake and strakes Plating between each pair of

STRUCTURE IN WING BALLAST TANKS

Structural member	Extent of measurement	Pattern of measurement
Side shell and longitudinal bulkhead plating:		Single measurement
- upper strake and strakes in way of horizontal girders	Plating between each pair of longitudinals in a minimum of three bays (along the tank)	
- all other strakes	Plating between every third pair of longitudinals in same three bays	Single measurement
Side shell and longitudinal bulkhead longitudinals on:	Each longitudinal in same three bays and I	3 measurements across web measurement on flange
- upper strake	Every third longitudinal in same three bays and I	3 measurements across web measurement on flange
Longitudina ls - brackets	Minimum of three at top, middle and bottom of tank in same three bays	5-point pattern over area of bracket
Vertical web and transverse bulkheads (excluding both transverse bulkheads deckhead area):	Minimum of two webs and two measurements between both transverse bulkheads	5-point pattern over approx. two square metre area each pair of vertical stiffeners
- strakes in way of horizontal girders		
- other strakes		

deckhead area): Minimum of two webs and two measurements between
- strakes in way of horizontal both transverse bulkheads each pair of vertical stiffeners
girders
- other strakes
Horizontal girders Plating on each girder in a Two measurements between
minimum of three bays each pair of longitudinal
girder stiffeners
Panel stiffening Where applicable Single measurements
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ANNEX 4/Sheet 4
REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT
AREAS OF SUBSTANTIAL CORROSION RENEWAL SURVEY OF DOUBLE
HULL OIL TANKERS WITHIN THE CARGO AREA LENGTH
LONGITUDINAL BULKHEADS IN CARGO TANKS
Structural member Extent of measurement Pattern of measurement
Deckhead and bottom strakes, Plating between each pair of Single measurement
and strakes in way of the longitudinals in a minimum of
horizontal stringers of three bays
transverse bulkhead s
All other strakes Plating between every third Single measurement
pair of longitudinals in same
three bays

and strakes in way of the longitudinals in a minimum of
horizontal stringers of three bays
transverse bulkhead s
All other strakes Plating between every third Single measurement
pair of longitudinals in same
three bays
Longitudinals on deckhead Each longitudinal in same Three measurements across
and bottom strakes three bays web and one measurement on
flange
All other longitudinals Every third longitudinal in Three measurements across
same three bays web and one measurement on
flange
Longitudinals - brackets Minimum of three at top, 5-point pattern over area of
middle and bottom of tank in bracket
same three bays
Web frames and cross ties Three webs with minimum of 5-point pattern over
three locations on each web, approximately two square
including in way of cross tie metre area of webs, plus
connections single measurements on
flanges of web frame and
cross ties
Lower end brackets Minimum of three brackets 5-point pattern over
(opposite side of web approximately two square
frame) metre area of brackets, plus
single measurements on
bracket flanges
RESOLUTION MSC.197(80)
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AMENDMENTS TO THE GUIDELINES ON THE ENHANCED

(opposite side of web approximately two square
frame) metre area of brackets, plus
single measurements on
bracket flanges

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ANNEX 4/Sheet 5
REQUIREMENTS FOR EXTENT OF THICKNESS MEASUREMENTS AT AREAS OF SUBSTANTIAL CORROSION RENEWAL SURVEY OF DOUBLE HULL OIL TANKERS WITHIN THE CARGO AREA LENGTH
TRANSVERSE WATERTIGHT AND SWASH BULKHEADS IN CARGO TANKS

Structural member	Extent of measurement	Pattern of measurement
Upper and lower stool, where fitted	Transverse band within 25 mm of welded connection to inner bottom/deck plating and to shelf plate	5-point pattern between stiffeners over one metre length
Deckhead and bottom strakes, and strak es in way of horizontal stringers	Plating between pair of stiffeners at three locations: approximately 1/4, 1/2 and 3/4 width of tank	5-point pattern between stiffeners over one metre length
All other strakes	Plating between pair of	Single measurement

and strak es in way of stiffeners at three locations: stiffeners over one metre
horizontal stringers approximately 1/4, 1/2 and 3/4 length
width of tank
All other strakes Plating between pair of Single measurement
stiffeners at middle location
Strakes in corrugated Plating for each change of 5-point pattern over about one
bulkheads scantling at centre of panel square metre of plating
and at flange of fabricated
connection
Stiffeners Minimum of three typical For web, 5-point pattern over
stiffeners span between bracket
connections (two
measurements across web at
each bracket connection and
one at centre of span).
For flange, single
measurements at each bracket
toe and at centre of span
Bracket s Minimum of three at top, 5-point pattern over area of
middle and bottom of tank bracket
Horizontal stringers All stringers with 5-point pattern over one
measurements at both ends square metre area, plus single
and middle measurement s near bracket
toes and on flanges

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and middle measurement s near bracket
toes and on flanges
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ANNEX 5
MINIMUM REQUIREMENTS FOR OVERALL AND CLOSE-UP SURVEY AND
THICKNESS MEASUREMENTS AT INTERMEDIATE SURVEY OF DOUBLE
HULL OIL TANKERS
5<age< 10 years 1 O<age< 15 years Age > 15 years
1 2 3
Overall survey of Overall survey of all As for renewal survey given
representative salt water salt water ballast tanks, in annex 1
ballast tanks, selected by the including combined
attending surveyor (the cargolballast tanks
selection should include fore where fitted (see 4.3)
and aft peak tanks and three
other tanks) (see 4.2)
Overall survey of at least two As for renewal survey given
representative cargo tanks in annex 1
Close-up survey in salt water As for renewal survey given
ballast tanks of: in annex I
- all web frames (1) in one
complete tank (see Note 1) -
the knuckle area and the
upper part (5 m
approximately) of one web
frame in each remaining
ballast tank (6)
- one transverse bulkhead (4)

ballast tanks of: in annex I
- all web frames (1) in one
complete tank (see Note 1) -
the knuckle area and the
upper part (5 m
approximately) of one web
frame in each remaining
ballast tank (6)
- one transverse bulkhead (4)
in each complete tank (see
Note 1) (see 4.2.3)
Close-up survey in two cargo As for renewal survey given
tanks (or two combined in annex I
cargolballast tanks, where
fitted):
the extent of survey should be
based on the records of the
previous renewal survey and
repair history of the tanks
(see 4.3)
Thickness measurements of Thickness measurements of As for renewal survey given
those areas found to be those areas found to be in annex 2
suspect areas, as defined in suspect areas, as defined in
1.2.7, at the previous renewal 1.2.7, at the previous renewal
survey (see 4.3.5) survey (see 4.3.5)
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Notes:
(1), (4) and (6) are areas to be subjected to close-up surveys and thickness measurements (see
appendix 3 of annex 3).

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AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 54-

Notes:

(1), (4) and (6) are areas to be subjected to close-up surveys and thickness measurements (see appendix 3 of annex 3).

(1) Web frame means vertical web in side tank, hopper web in hopper tank, floor in double bottom tank and deck transverse in double deck tank (where fitted), including adjacent structural members. In fore and aft peak tanks web frame means a complete transverse web frame ring including adjacent structural members.

(4) Transverse bulkhead complete in ballast tanks, including girder system and adjacent structural members, such as longitudinal bulkheads, girders in double bottom tanks, inner bottom plating, hopper side, inner hull longitudinal bulkhead, connecting brackets.

(6) The knuckle area and the upper part (5 metres approximately), including adjacent structural members. Knuckle area is the area of the web frame around the connections of the slope hopper plating to the inner hull bulkhead and the inner bottom plating, up to 2 metres from the comers both on the bulkhead and the double bottom.

structural members. Knuckle area is the area of the web frame around the connections of the slope hopper plating to the inner hull bulkhead and the inner bottom plating, up to 2 metres from the comers both on the bulkhead and the double bottom.

Note 1 Complete ballast tank: means double bottom tank plus double side tank plus double deck tank, as applicable, even if these tanks are separate.

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ANNEX6A

SURVEY PROGRAMME

Basic information and particulars

Name of ship:

IMO number:

Flag State:

Port of registry :

Gross tonnage:

Deadweight (metric tonnes) :

Length between perpendiculars (m) :

Shipbuilder:

Hull number:

Recognized organization (RO) :

RO ship identity :

Date of delivery of the ship:

Owner :

Thickness measurement company:

1 Preamble

1.1 Scope

1.1.1 The present survey programme covers the minimum extent of overall surveys, close-up surveys, thickness measurements and pressure testing within the cargo area,

Owner :
Thickness measurement company:
1 Preamble
1.1 Scope
1.1.1 The present survey programme covers the minimum extent of overall surveys, close-up surveys, thickness measurements and pressure testing within the cargo area, ballast tanks, including fore and aft peak tanks, required by the Guidelines.
1.1.2 The arrangements and safety aspects of the survey should be acceptable to the attending surveyor(s).
1.2 Documentation
All documents used in the development of the survey programme should be available on board during the survey as required by section 6.
2 Arrangement of tanks and spaces
This section of the survey programme should provide information (either in the form of plans or text) on the arrangement of tanks and spaces that fall within the scope of the survey.
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3 List of tanks and spaces with information on their use, extent of coatings and corrosion protection system

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3 List of tanks and spaces with information on their use, extent of coatings and corrosion protection system

This section of the survey programme should indicate any changes relating to (and should update) the information on the use of the tanks of the ship, the extent of coatings and the corrosion protective system provided in the Survey Planning Questionnaire.

4 Conditions for survey

This section of the survey programme should provide information on the conditions for survey, e.g. information regarding cargo hold and tank cleaning, gas freeing, ventilation, lighting, etc.

5 Provisions and method of access to structures

This section of the survey programme should indicate any changes relating to (and should update) the information on the provisions and methods of access to structures provided in the Survey Planning Questionnaire.

6 List of equipment for survey

This section of the survey programme should identifY and list the equipment that will be made available for carrying out the survey and the required thickness measurements.

7 Survey requirements

6 List of equipment for survey
This section of the survey programme should identifY and list the equipment that will be made available for carrying out the survey and the required thickness measurements.
7 Survey requirements
7.1 Overall survey
This section of the survey programme should identifY and list the spaces that should undergo an overall survey for the ship in accordance with 2.4.1.
7.2 Close-up survey
This section of the survey programme should identifY and list the hull structures that should undergo a close-up survey for the ship in accordance with 2.4.2.
8 Identification of tanks for tank testing
This section of the survey programme should identifY and list the tanks that should undergo tank testing for the ship in accordance with 2.6.
9 Identification of areas and sections for thickness measurements
This section of the survey programme should identifY and list the areas and sections where thickness measurements should be taken in accordance with 2.5.1.
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where thickness measurements should be taken in accordance with 2.5.1.
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10 Minimum thickness of hull structures
This section of the survey programme should specifY the minimum thickness for hull structures of the ship that are subject to the Guidelines (indicate either (a) or preferably (b), if such information is available):
(a) o Determined from the attached wastage allowance table and the original thickness to the hull structure plans of the ship;
(b) o Given in the following table(s):
Original as- Minimum Substantial corrosion Area or location built thickness thickness thickness (mm) (mm) (mm)
Deck
Plating
Longitudinals
Longitudinal girders
Bottom
Plating
Longitudinals
Longitudinal girders
Ship side
Plating
Longitudinals
Longitudinal girders
Longitudinal bulkhead
Plating
Longitudinals
Longitudinal girders
Inner bottom
Plating
Longitudinals
Longitudinal girders
Transverse bulkheads
Plating
Stiffeners
Transverse web frames, floors

Plating
Longitudinals
Longitudinal girders
Longitudinal bulkhead
Plating
Longitudinals
Longitudinal girders
Inner bottom
Plating
Longitudinals
Longitudinal girders
Transverse bulkheads
Plating
Stiffeners
Transverse web frames, floors
and stringers
Plating
Flanges
Stiffeners
Cross ties
Flanges
Webs
Note: The wastage allowance tables should be attached to the survey programme.
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11 Thickness measurement company
This section of the survey programme should identifY changes, if any, relating to the information on the thickness measurement company provided in the Survey Planning Questionnaire.
12 Damage experience related to the ship
This section of the survey programme should, using the tables provided below, provide details of the hull damages for at least the last three years in way of the cargo and ballast tanks and void spaces within the cargo area. These damages are subject to survey.
Hull damages sorted by location for the ship

details of the hull damages for at least the last three years in way of the cargo and ballast tanks and void spaces within the cargo area. These damages are subject to survey.

Hull damages sorted by location for the ship

Tank or Possible Description space of Location Repair Date of number or cause, repair if known the damages area

Hull damages for sister or similar ships (if available) in the case of design related damage

Tank or Possible Description space of Location Repair Date of number or cause, repair if known the damages area

13 Areas identified with substantial corrosion from previous surveys

This section of the survey programme should identify and list the areas of substantial corrosion from previous surveys.

14 Critical structural areas and suspect areas

This section of the survey programme should identify and list the critical structural areas and the suspect areas, if such information is available.

15 Other relevant comments and information

14 Critical structural areas and suspect areas
This section of the survey programme should identify and list the critical structural areas and the suspect areas, if such information is available.
15 Other relevant comments and information
This section of the survey programme should provide any other comments and information relevant to the survey.
Appendices
Appendix 1 - List of plans
5.1.3.2 requires that main structural plans of cargo and ballast tanks (scantling drawings), including information on regarding use of high tensile steel (HTS), to be available. This appendix of the survey programme should identify and list the main structural plans which form part of the survey programme.
Appendix 2 - Survey Planning Questionnaire
The Survey Planning Questionnaire (annex 6B), which has been submitted by the owner, should be appended to the survey programme.
Appendix 3 - Other documentation
This part of the survey programme should identify and list any other documentation that forms part of the Plan.
Prepared by the owner in co-operation with the Administration for compliance with 5.1.3.
Date: ..... ".
Date: ... ,," " .. " .....

forms part of the Plan.
Prepared by the owner in co-operation with the Administration for compliance with 5.1.3.
Date: ..... ".
Date: ... ,," " .. " .....
(name and signature of authorized owner's representative)
(name and signature of authorized representative of the
Administration)
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ANNEX6B
SURVEY PLANNING QUESTIONNAIRE
The following information will enable the owner in co-operation with the Administration to develop a survey programme complying with the requirements of the Guidelines . It is essential that the owner provides , when completing the present questionnaire , up-to-date information. The present questionnaire, when completed, should provide all information and material required by the resolution.
Particulars
Ship's name:
IMO number:
Flag State:
Port of registry:
Owner:
Recognized organization (RO):
Gross tonnage:
Deadweight (metric tonnes):
Date of delivery:

and material required by the resolution.
Particulars
Ship's name:
IMO number:
Flag State:
Port of registry:
Owner:
Recognized organization (RO):
Gross tonnage:
Deadweight (metric tonnes):
Date of delivery:
Information on access provision for close-up surveys and thickness measurement:
The owner should indicate, in the table below, the means of access to the structures subject to close-up survey and thickness measurement. A close-up survey is an examination where the details of structural components are within the close visual inspection range of the attending surveyor, i.e. preferably within reach of hand.
Ta nk Structure C (Cargo) / Temporary Rafts Ladders Direct Other
No. B (Ballast) staging access means
(please
specify)
F.P. Fore peak
A.P. Aftpeak
Under-deck
Side shell
~ ..., Bottom c
'" transverse -bll Longitudinal = ~ Transverse
Under-deck
.. Bottom ... ~
-..., transverse = = Ut! Transverse
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-..., transverse = = Ut! Transverse
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History of cargo with HzS content or heated cargo for the last 3 years together with indication as to whether cargo was heated and, where available, Marine Safety Data Sheets (MSDS)
Owner's inspections
Using a fonnat similar to th at of the table below (which is given as an example), the owner should provide details of the results of their inspections for the last 3 years on all cargo and ballast tanks and void spaces within the cargo area, including peak tanks.

Tank No.	Corrosion protection	Coating extent	Coating condition	Structural deterioration	Tank damage history
(1)	(2)	(3)	(4)	(5)	
Cargo centre tanks					
Cargo wing tanks					
Slop					

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Tank No.	Corrosion protection	Coating extent
(1)	(2)	
Banast tanks		
Aft peak		
Fore peak		

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Tank No.	Corrosion protection	Coating extent	Coating condition	Structural deterioration	Tank damage history
(1)	(2)			(3)	(4) (5)
Banast tanks					
Aft peak					
Fore peak					
Miscellaneous spaces					

Note: Indicate tanks which are used for oillballast.

I) HC=hard coating; SC=soft coating; A=anodes ; NP=no protection

2) U=upper part; M=middle part; L=lower part; C=complete

3) G=good; F=fair; P=poor; RC=recoated (during the last 3 years)

4) N=no findings recorded ; Y=findings recorded , description of findings should be attached to the questionnaire

5) DR=damage and repair ; L=leakage s; CY= conversion (description should be attached to this questionnaire)

Reports of Port State Control inspections

Name of owner's repre sentative :

Signature: ....

Date: ....

List the reports of Port State Control inspections containing hull structural-related deficiencies and relevant information on rectification of the deficiencies:

Date: ....
List the reports of Port State Control inspections containing hull structural-related deficiencies and relevant information on rectification of the deficiencies:
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Safety Management System
List of non-conformities related to hull maintenance , including the associated corrective actions:
Name and address of the approved thickness measurement company:
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ANNEX6C
OWNER'S INSPECTION REPORT
Structural condition
Ship's name: ....
For tank No.: ....
Grade of steel: deck: side: ....

Ship's name: ....
For tank No.: ....
Grade of steel: deck: side: ....
bottom: .... longitudinal bulkhead: ..... .
Coating Modifi-Elements Cracks Buckles Corrosion Pitting cation! Other condition repair
Deck
Bottom
Side
Longitudinal
bulkhead
Transverse
bulkhead
Repairs carried out due to:
Thickness measurements carried out (dates):
Results in general:
Overdue surveys:
Outstanding conditions of class:
Comments:
Date of inspection:
Inspected by:
Signature:
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ANNEX 7
PROCEDURES FOR CERTIFICATION OF A COMPANY ENGAGED IN
THICKNESS MEASUREMENT OF HULL STRUCTURES
1 Application
This guidance applies for certification of the company which intends to engage in the thickness measurement of hull structures of ships.

PROCEDURES FOR CERTIFICATION OF A COMPANY ENGAGED IN THICKNESS MEASUREMENT OF HULL STRUCTURES

1 Application

This guidance applies for certification of the company which intends to engage in the thickness measurement of hull structures of ships.

2 Procedures for certification

Submission of documents

2.1 The following documents should be submitted to an organization recognized by the Administration for approval:

.1 outline of the company, e.g. organization and management structure;

.2 experience of the company in thickness measurement of hull structures of ships;

.3 technicians' careers, i.e. experience of technicians as thickness measurement operators, technical knowledge and experience of hull structure, etc. Operators should be qualified according to a recognized industrial non-destructive test (NDT) standard;

.4 equipment used for thickness measurement such as ultrasonic testing machines and their maintenance/calibration procedures;

.5 a guide for thickness measurement operators;

.6 training programmes for technicians for thickness measurement; and

.7 measurement record format in accordance with recommended procedures

machines and their maintenance/calibration procedures;
.5 a guide for thickness measurement operators;
.6 training programmes for technicians for thickness measurement; and
.7 measurement record format in accordance with recommended procedures for thickness measurements (see annex 10). Recommended procedures for thickness measurements of double hull oil tankers are contained in annex 2.

Auditing of the company

2.2 Upon satisfactory review of the documents submitted, the company should be audited in order to ascertain that the company is duly organized and managed in accordance with the documents submitted and is capable of conducting thickness measurement of the hull structure of ships.

2.3 Certification is conditional upon an on-board demonstration of thickness measurement as well as satisfactory reporting.

3 Certification

3.1 Upon satisfactory results of both the audit of the company referred to in 2.2 and the demonstration tests referred to in 2.3, the Administration or organization recognized by the Administration should issue a certificate of approval as well as a notice to the effect that the thickness measurement operation system of the company has been

by the Administration should issue a certificate of approval as well as a notice to the effect that the thickness measurement operation system of the company has been certified.

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3.2 Renewal/endorsement of the certificate should be carried out at intervals not exceeding three years by verification that original conditions are maintained.

4 Report of any alteration to the certified thickness measurement operation system

In case any alteration to the certified thickness measurement operation system of the company is made, such alteration should be immediately reported to the organization recognized by the Administration. A re-audit should be carried out where deemed necessary by the organization recognized by the Administration.

5 Withdrawal of certification

The certification may be withdrawn in the following cases:

.1 where the measurements were improperly carried out or the results were improperly reported;

5 Withdrawal of certification
The certification may be withdrawn in the following cases:
.1 where the measurements were improperly carried out or the results were improperly reported;
.2 where the surveyor found any deficiencies in the approved thickness measurement operation system of the company;
.3 where the company failed to report any alteration referred to in 4 to the organization recognized by the Administration as required.
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ANNEX 8
SURVEY REPORTING PRINCIPLES
As a principle, for oil tankers subject to the Guidelines, the surveyor should include the following contents in his report for survey of hull structure and piping systems, as relevant for the survey.
1 General
1.1 A survey report should be generated in the following cases:
. I in connection with commencement , continuation and/or completion of periodical hull surveys, i.e. annual, intermediate and renewal surveys, as relevant;
.2 when structural damages/defects have been found;

. I in connection with commencement , continuation and/or completion of periodical hull surveys, i.e. annual, intermediate and renewal surveys, as relevant;
.2 when structural damages/defects have been found;
.3 when repairs, renewals or modifications have been carried out; and
.4 when condition of class (recommendation) has been imposed or deleted.
1.2 The reporting should provide:
.1 evidence that prescribed surveys have been carried out in accordance with applicable requirements;
.2 documentation of surveys carried out with findings, repairs carried out and condition of class (recommendation) imposed or deleted;
.3 survey record s, including actions taken, which should form an auditable documentary trail. Survey reports should be kept in the survey report file required to be on board;
.4 information for planning of future surveys; and
.5 information which may be used as input for maintenance of classification rules and instructions .
1.3 When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items surveyed, relevant findings and an

rules and instructions .
1.3 When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items surveyed, relevant findings and an indication of whether the item has been credited , should be made available to the next attending surveyor, prior to continuing or completing the survey. Thickness measurement and tank testing carried out should also be listed for the next surveyor.
2 Extent of the survey
2.1 Identification of compartments where an overall survey has been carried out.
2.2 Identification of locations, in each tank, where a close-up survey has been carried out, together with information of the means of access used.
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2.3 Identification of locations, in each tank, where thickness measurement has been carried out.
Note: As a minimum, the identification of location of close-up survey and thickness measurement should include a confirmation with description of individual

carried out.
Note: As a minimum, the identification of location of close-up survey and thickness measurement should include a confirmation with description of individual structural members corresponding to the extent of requirements stipulated in this part of Annex B based on type of periodical survey and the ship's age.
Where only partial survey is required, i.e. one web frame ring/one deck transverse, the identification should include location within each ballast tank and cargo hold by reference to frame numbers.
2.4 For areas in tanks where protective coating is found to be in good condition and the extent of close-up survey and/or thickness measurement has been specially considered , structures subject to special consideration should be identified.
2.5 Identification of tanks subject to tank testing.
2.6 Identification of piping systems on deck, including crude oil washing (COW) piping , and ballast piping within cargo and ballast tanks, pipe tunnels, cofferdams and void spaces where:
.1 examination including internal examination of piping with valves and fittings and thickness measurement, as relevant, has been carried out; and

void spaces where:
.1 examination including internal examination of piping with valves and fittings and thickness measurement, as relevant, has been carried out; and
.2 operational test to working pressure has been carried out.
3 Result of survey
3.1 Type, extent and condition of protective coating in each tank, as relevant (rated GOOD, FAIR or POOR), including identification of tanks fitted with anodes.
3.2 Structural condition of each compartment with information on the following, as relevant:
.1 Identification of findings, such as:
.1.1 corrosion with description oflocation, type and extent;
.1.2 areas with substantial corrosion ;
.1 .3 cracks/fractures with description of location and extent;
.1.4 buckling with description ofloca tion and extent; and
.1.5 indents with description of location and extent.
.2 Identification of compartments where no structural damages/defects are found. The report may be supplemented by sketches/photos .
. 3 Thickness measurement report should be verified and signed by the surveyor controlling the measurements on board.
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. 3 Thickness measurement report should be verified and signed by the surveyor controlling the measurements on board.

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.4 Evaluation result of longitudinal strength of the hull girder of oil tankers of 130 m in length and upwards and over 10 years of age. The following data should be included, as relevant:

.4.1 measured and as-built transverse sectional areas of deck and bottom flanges;

.4.2 diminution of transverse sectional areas of deck and bottom flanges; and

.4.3 details of renewals or reinforcements carried out, as relevant (see 4.2).

4 Actions taken with respect to findings

4.1 Whenever the attending surveyor is of the opinion that repairs are required, each item to be repaired should be identified in a numbered list. Whenever repairs are carried out, details of the repairs effected should be reported by making specific reference to relevant items in the numbered list.

4.2 Repairs carried out should be reported with identification of:

.1 compartment;

out, details of the repairs effected should be reported by making specific reference to relevant items in the numbered list.
4.2 Repairs carried out should be reported with identification of:
.1 compartment;
.2 structural member;
.3 repair method (i.e. renewal or modification), including:
.3.1 steel grades and scantlings (if different from the original);
.3.2 sketches/photos, as appropriate;
.4 repair extent; and
.5 non-destructive tests (NDT).
4.3 For repairs not completed at the time of survey, condition of class/recommendation should be imposed with a specific time limit for the repairs. In order to provide correct and proper information to the surveyor attending for survey of the repairs, condition of class/recommendation should be sufficiently detailed with identification of each item to be repaired. For identification of extensive repairs, reference may be given to the survey report.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)General particulars
Ship's name:
Port of registry :
Deadweight

(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)General particulars
Ship's name:
Port of registry :
Deadweight
(metric tonnes):
Date of build :
- 70-
ANNEX 9
CONDITION EVALUATION REPORT
Issued upon completion of renewal survey
Class/Administration identity number:
Previous class/Administration identity number(s):
IMOnumber :
National flag:
Previou s national flag(s):
Gross tonnage:
National:
ITC (1969):
Classification notation:
Date of major conversion :
Type of conversion: Owner:
Previous owner(s):
1 The survey reports and documents listed below have been reviewed by the
undersigned and found to be satisfactory.
2 The renewal survey has been completed in accordance with the present
Guideline s on (date) ....
Condition evaluation report Name Title
completed by Signature
Office Date
Condition evaluation report Name Title
completed by Signature
Office Date
Attached report s and documents:
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED

completed by Signature
Office Date
Condition evaluation report Name Title
completed by Signature
Office Date
Attached report s and documents:
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 71 -
Contents of condition evaluation report
Part 1 - General particulars:
Part 2 - Report review:
Part 3 - Close-up survey:
Part 4 - Cargo and ballast
piping system:
Part 5 - Thickness measurements:
Part 6 - Tank corrosion
prevention system:
Part 7 - Repairs:
Part 8 - Condition of class/
flag State requirements:
Part 9 - Memoranda:
Part 10 - Conclusion:
See front page
Where and how survey was done
Extent (which tanks)
Examined
Operationally tested
Reference to thickness measurement report
Summary of where measured
Separate form indicating the spaces with
substantial corrosion, and corresponding:
- thickness diminution
- corrosion pattern
Separate form indicating:
- location of coating/anodes
- condition of coating (if applicable)
Identification of tanks/areas
- Acceptable defects

substantial corrosion, and corresponding:
- thickness diminution
- corrosion pattern
Separate form indicating:
- location of coating/anodes
- condition of coating (if applicable)
Identification of tanks/areas
- Acceptable defects
- Any points of attention for future surveys,
e.g. for suspect areas
Extended annuaVintermediate survey due to
coating breakdown
Statement on evaluation/verification of
survey report
Extract of thickness measurements
Reference is made to the thickness measurements report :
Position of substantiaUy Thickness Corros ion Remarks: corroded tanks /areas! or (e.g. reference to
areas with deep pitting3 diminution (%) pattern' attached sketches)
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)-72-
Notes:
I Substantial corrosion, i.e. 75%-100% of acceptable margins wasted.
2 P = Pitting
C Corrosion in general
3 Any bottom plating with a pitting intensity of 20% or more, with wastage in the
substantial corrosion range or having an average depth of pitting of 1/3 or more of

2 P = Pitting
C Corrosion in general
3 Any bottom plating with a pitting intensity of 20% or more, with wastage in the substantial corrosion range or having an average depth of pitting of 1/3 or more of actual plate thicknes s should be noted .
Tank corrosion prevention system
Tank Nos.' Tank corrosion Coating condition' Remarks
prevention system2
Notes:
All segregated ballast tanks and combined cargolballast tanks should be listed.
2 C = Coating
A = Anodes
NP = No protection
3 Coating condition according to the following standard:
GOOD
FAIR
POOR
condition with only minor spot rusting.
condition with local breakdown of coating at edges of stiffeners and weld connections and/or light rusting over 20% or more of areas under consideration, but less than as defined for POOR condition.
condition with general breakdown of coating over 20% or more of areas or hard scale at 10% or more of areas under consideration.
If coating condition "POOR " is given, extended annual surveys should be introduced .
This should be noted in part 8 of the contents of the condition evaluation report .

or hard scale at 10% or more of areas under consideration.
If coating condition "POOR " is given, extended annual surveys should be introduced .
This should be noted in part 8 of the contents of the condition evaluation report .
Evaluation result oflongitudinal strength of the hull girder of oil tankers of 130 m
in length and upwards and of over 10 years of age
(of sections 1,2 and 3 below, only one applicable section should be completed)
I This section applies to ships regardless of the date of construction: Transverse
sectional areas of deck flange (deck plating and deck longitudinals) and bottom flange
(bottom shell plating and bottom longitudinals) of the ship's hull girder have been
calculated by using the thickness measured, renewed or reinforced , as appropriate, during
the renewal survey of the Cargo Ship Safety Construction Certificate or the Cargo Ship
Safety Certificate (SC renewal survey) most recently conducted after the ship reached
10 years of age, and found that the diminution of the transverse sectional area does not
exceed 10% ofthe as-built area, as shown in the following table:
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)

10 years of age, and found that the diminution of the transverse sectional area does not exceed 10% ofthe as-built area, as shown in the following table:
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 73 -
Table 1 - Transverse sectional area of hull girder flange
Measured As-built Diminution
Transverse Deek flange em' em' em'(%)
seetion I Bottom flange em' em' em'(%)
Transverse Deek flange em' em' em' (%)
seetion 2 Bottom Flange em' em' em'(%)
Transverse Deek Flange em' em' em' (%)
seetion 3 Bottom Flange em' em' em'(%)
2 This section applies to ships constructed on or after I July 2002: Section moduli of transverse section of the ship 's hull girder have been calculated by using the thickness of structural members measured, renewed or reinforced , as appropriate, during the SC renewal survey most recently conducted after the ship reached 10 years of age in accordance with the provision s of paragraph 2.1.1 of annex 12, and are found to be

renewal survey most recently conducted after the ship reached 10 years of age in accordance with the provision s of paragraph 2.1.1 of annex 12, and are found to be within their diminution limits determined by the Administration, taking into account the recommendations adopted by the Organization, as shown in the following table:

Table 2 - Transverse section modulus of bull girder

	Z"" (em')	Zreq (em')'	Remarks
Transverse Upper deek seetion I Bottom			
Transverse Upper deek seetion 2 Bottom			
Transverse Upper deck section 3 Bottom			

Notes:

1 Zact means the actual section moduli of the transverse section of the ship ' s hull girder calculated by using the thickness of structural members measured, renewed or reinforced , as appropriate, during the SC renewal survey, in accordance with the provisions of paragraph 2. L I of annex 12.

2 Zreq means the diminution limit of the longitudinal bending strength of ships, as calculated in accordance with the provisions of paragraph 2.1.1 of annex 12.

The calculation sheets for Z ac! should be attached to this report .

3 This section applies to ships constructed before I July 2002: Section moduli of

ships, as calculated in accordance with the provisions of paragraph 2.1.1
of annex 12.
The calculation sheets for Z ac! should be attached to this report .
3 This section applies to ships constructed before I July 2002: Section moduli of
transverse sections of the ship's hull girder have been calculated by using the thicknes s of
structural members measured , renewed or reinforced, as appropriate, during the
SC renewal survey most recentl y conducted after the ship reached 10 years of age in
accordance with the provision s of paragraph 2.2.1.2 of annex 12, and found to meet the
criteria required by the Administration or the recogni zed classification society and that
Zact is not less than Zmc (defined in Note 2 below) as specified in appendix 2 to annex 12,
as shown in the following table.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 74-
Describe the criteria for acceptance of the minimum section moduli of the ship's hull

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 74-

Describe the criteria for acceptance of the minimum section moduli of the ship's hull girder for ships in service required by the Administration or the recognized classification society.

Table 3 - Transverse section modulus of hull girder

Zoo, (em')'	Zm, (cm')"	Remarks
Transverse section I	Upper deck Bottom	
Transverse section 2	Upper deck Bottom	
Transverse section 3	Upper deck Bottom	

Notes:

I As defined in note I of table 2.

2 Zm< means the diminution limit of minimum section modulus calculated in accordance with provisions of paragraph 2.1.2 of annex 12.

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(adopted on 20 May 2005)

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 75 -

ANNEX 10

RECOMMENDED PROCEDURES FOR THICKNESS MEASUREMENTS
OF DOUBLE HULL OIL TANKERS

General

1 These procedures should be used for recording thickness measurements as required by annexes 2 and 4.

ANNEX 10
RECOMMENDED PROCEDURES FOR THICKNESS MEASUREMENTS
OF DOUBLE HULL OIL TANKERS
General
1 These procedures should be used for recording thickness measurements as required by annexes 2 and 4.
2 Reporting fonns TMI-DHT , TM2-DHT(i), TM2-DHT(ii), TM3-DHT, TM4-DHT, TM5-DHT and TM6-DHT, set out in appendix 2, should be used for recording thickness measurements and the maximum allowable diminution should be stated.
3 Appendix 3 contains guidance diagrams and notes relating to the reporting fonns and the requirements for thickness measurement.
4 The reporting fonns should, where appropriate, be supplemented by data presented on structural sketches.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 76-
APPENDIXl
Ship's name: ....
IMO number: ..... .

IMO number: ..... .
Class/Administration identity number: ....
Port of registry: ....
Gross tonnage: ....
Deadweight: ....
Date of build: ....
Classification society: ....
N arne of company performing thickness measurement: ....
Thickness measurement company certified by: ..... .

N arne of company performing thickness measurement: ....
Thickness measurement company certified by: ..... .
Certificate number:
Certificate valid from: to
Place of measurement: ....
First date of measurement: ....
Last date of measurement:
Renewal survey/intermediate survey' due: ....
Details of measurement equipment: ....
Qualification of operator: ....
Report number: .... consisting of ..... pages.
Name of operator: .... Name of surveyor: ..... .

Report number: .... consisting of ..... pages.
Name of operator: .... Name of surveyor: ..... .
Signature of operator: .... Signature of surveyor: ....
Company official stamp: ..... Administration: ....
Official stamp: ..... .
Delete as appropriate .
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)STRAKE
POSITION
PLATE
POSITIO N
II '" for
10'"
9
t
"
8'"
7'"
6
th
5'"
4th
3'"
2""
I "
Amid ships
1 sl aft
2""
3'"
4th
5'"
6'"
7"
8'"
9'"
10
th
II '"
- 77-
APPENDIX 2
TMI-DHT Report on thickness measurement of aD deck plating, aD bottom sheD
plating or side sheD plating'
Ship's name .. _____ . __ .__ _____ Class identity No. ____ ____ ____ _____
_____ _

TMI-DHT Report on thickness measurement of aD deck plating, aD bottom sheD plating or side sheD plating'

Ship's name .. _____ . __ .__ _____ Class identity No. ____ ____ ____ _____ _____ _

Report No .. __ _____ _ . __ . _____ . __ . _____ _

No. Original Forward reading Aft readin
or thickness Gauged Diminution Diminution Gauged Diminution Diminution %
letter {mm} P S P S
P S mIll % mm % P S mm % mm % P S
mm

Operator's signature __ ____ __ _____ ___ ._____ ____ ___ Surveyor's signature _____ _ . ___ _____ . ___ _____ _

NOTES - See reverse

Delete as appropriate .

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(adopted on 20 May 2005)

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 78-

NOTES TO THE REPORT TMI-DHT:

This report should be used for recording the thickness measurement of:

.1 All strength deck plating within the cargo area .

. 2 All keel, bottom shell plating and bilge plating within the cargo area .

NOTES TO THE REPORT TMI-DHT:
This report should be used for recording the thickness measurement of:
.1 All strength deck plating within the cargo area .
. 2 All keel, bottom shell plating and bilge plating within the cargo area .
. 3 Side shell plating, including selected wind and water strakes outside cargo area .
.4 All wind and water strakes within cargo area.
2 The strake position should be clearly indicated as follows:
.1 For strength deck indicate the number of the strake of plating inboard from the stringer plate .
.2 For bottom plating indicate the number of the strake of plating outboard from the keel plate .
. 3 For side shell plating give number of the strake of plating below sheerstrake and letter as shown on shell expansion.
3 Measurements should be taken at the forward and aft areas of all and where plates cross ballast/cargo tank boundaries separate measurements for the area of plating in way of each type of tank should be recorded.
4 The single measurements recorded are to represent the average of multiple measurements.
4 The maximum allowable diminution could be stated in an attached document.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)

4 The single measurements recorded are to represent the average of multiple measurements.

4 The maximum allowable diminution could be stated in an attached document.

RESOLUTION MSC.197(80)

(adopted on 20 May 2005)

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED))

- 79-

TM2- DHT (i) Report on thickness measurement of shell and deck plating (one, two or three transverse sections)

Ship's name .... Class identity No. .... Report No ....

STRENGTH DECK AND SHEERSTRAKE PLATING

FIRST TRANSVERSE SECTION AT FRAME NUMBER SECOND TRANSVERSE SECTtON AT FRAME NUMBER THIRD TRANSVERSE SECTION AT FRAME NUMBER

STRAKE No. Original Maximum Gauged Diminution Diminution No. Original Maximum Gauged Diminution Diminution No. or Original Maximum Gauged Diminution Diminution

POStTtON or tltickre;.o; allowable P S or thickness allowable P S Letter thickness allowable P S

POStTtON or tltickre;.o; allowable P S or thickness allowable P S Letter thickness allowable P S
Letter diminution r..tter diminution diminution
mm mm P S mm % mm % mm mm P S mm % S % mm mm P S mm % mm %
Stringer
Plate
1st stra ke
inboard
2"
3
4
5
6
7
8
9
10
lHh
12"
13"
14"
Centre srrake
Sheer strake
TOPSIDE
TOTAL
Operator's Signature ..... , ... , .. Surveyor's Signature .... NOTES - See reverse
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 80-
NOTES TO THE REPORT TM2-DHT (i)
I This report form should be used for recording the thickness measurements of
strength deck plating and sheerstrake plating transverse sections:
One, two or three sections within the cargo area comprising of the structural items
(0), (I) and (2) as shown on the diagrams of typical transverse sections illustrated
in appendix 3 of annex 10.
2 The topside area comprises deck plating, stringer plate and sheerstrake (including
rounded gunwales).

(0), (I) and (2) as shown on the diagrams of typical transverse sections illustrated in appendix 3 of annex 10.
2 The topside area comprises deck plating, stringer plate and sheerstrake (including rounded gunwales).
3 The exact frame station of measurement should be stated.
4 The single measurements recorded are to represent the average of multiple measurements.
5 The maximum allowable diminution could be stated in an attached document.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)))
- 81 -
TM2--DHT (ii) Report on thickness measurement of shell and deck plating (one, two or three transverse sections)
Ship's name .... Class identity No. ..... Report No ....
SHELL PLATING
FIRST TRANSVERSE SECTION AT FRAME NUMBER SECOND TRANSVERSE SECTION AT FRAME NUMBER THIRD TRANSVERSE SECTION AT FRAME NUMBER

SHELL PLATING
FIRST TRANSVERSE SECTION AT FRAME NUMBER SECOND TRANSVERSE SECTION AT FRAME NUMBER THIRD TRANSVERSE SECTION AT FRAME NUMBER
STRAKE No. Origiool Maximum Gauged Diminution Diminution No. Original Maximum Gauged Diminution Diminution No. or Original Maximum Gauged Diminution Diminution
POSmON or .hid",,,,, allowable P S or thickness allowable P S Letter thickness allowable P S
Le"", diminution Letter diminution diminution
mm mm P S mm % mm % mm mm P S mm % S % mm mm P S mm % mm %
lsrbelow
sheer strake
2
3'
4
5
6
7
S
9
10"
17"
IS"
19"
20
Keel strake
Bottom total
Operator 's Signature ..... . Surveyor's Signature .... , .. , ... NOTES - See reverse
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 82-
NOTES TO THE REPORT TM2-DHT(ii)
1 This report form should be used for recording the thickness measurements of shell
plating transverse sections:
One, two or three sections within the cargo area comprising ofthe structural items

NOTES TO THE REPORT TM2-DHT(ii)

1 This report form should be used for recording the thickness measurements of shell plating transverse sections:

One, two or three sections within the cargo area comprising ofthe structural items (3), (4) and (5) and (6) as shown on the diagrams of typical transverse sections illustrated in appendix 3 of annex 10.

2 The bottom area comprises keel, bottom and bilge plating.

3 The exact frame station of measurement should be stated.

4 The single measurements recorded are to represent the average of multiple measurements.

5 The maximum allowable diminution could be stated in an attached document. .

RESOLUTION MSC.197(80)

(adopted on 20 May 2005)

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)))

TM3-DHT Report on thickness measurement of longitudinal members (one, two or three transverse sections)

PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)))
- 83 -
TM3-DHT Report on thickness measurement of longitudinal members (one, two or three transverse sections)
Ship's name _____ _____ _____ _____ . Class identity N 0 _____
_____ ___ _____ . Report No. _____
_____ __
FIRST TRANSVERSE SECTION AT FRAME NUMBER SECOND TRANSVERSE SECTrON AT FRAME NUMBER THIRD TRANSVERSE SECTION AT FRAME NUMBER
Structural Item Original Maximum Gauged Diminution Diminution Item Original Maximum Gauged Diminution Diminution Item Original Maximum Gauged Diminution Diminution
member No. thickness allowable P S No. thickness allowable P S No. thickness allowable P S
diminution diminution diminution
mm mm P S mm % mm % mm mm P S mm % S % mm mm P S mm % mm %
Operator's Signature .... Surveyor's Signature .... NOTES - See reverse
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED

Operator's Signature .... Surveyor's Signature .... NOTES - See reverse
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 84-
NOTES TO THE REPORT TM3-DHT
I This report should be used for recording the thickness measurements of longitudinal members at transverse sections:
One, two or three sections within the cargo area comprising of the appropriate structural items (10) to (29) as shown on the diagrams of typical transverse sections illustrated in appendix 3 of annex 10.
2 The exact frame station of measurement should be stated.
3 The single measurements recorded are to represent the average of multiple measurements.
4 The maximum allowable diminution could be stated in an attached document.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED))

RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED))
- 85 -
TM4-DHT Report on thickness measurement of transverse structural members
in the cargo oil and water ballast tanks within the cargo tauk leugth
Ship' s name..... Class identity No. ..... Report No ..... .
TANK DESCRIPTION
LOCATION OF STRUCTURE
STRUCTURAL ITEM Original Maximum Gauged Diminution Diminution
MEMBER thickness allowable P S
(mm) diminution P S mm % mm %
(mm)
Operator's Signature .... Surveyor's Signature .... NOTES - See reverse
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 86-
NOTES TO THE REPORT TM4-DHT
I This report should be used for recording the thickness measurements of transverse

PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 86-

NOTES TO THE REPORT TM4-DHT

I This report should be used for recording the thickness measurements of transverse structural members, comprising of the appropriate structural items (30) to (36) as shown on diagrams of typical transverse sections illustrated in appendix 3 of annex 10.

2 Guidance for areas of measurement is indicated in appendix 3 of annex 10. The single measurements recorded are to represent the average of multiple measurements.

3 The maximum allowable diminution could be stated in an attached document.

RESOLUTION MSC.197(80)

(adopted on 20 May 2005)

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)TMS-DHT)

- 87-

Report on thickness ofW.T'!O.T transverse bulkheads within the cargo tank or cargo hold spaces

Ship's name _____ Class identity No_ __ _____ _____ Report No _____ _

TANKlHOLD Description

LOCATION OF STRUCTURE FRAME No

Ship's name _____ Class identity No_ __ _____ _____ Report No _____ _

TANKlHOLD Description

LOCATION OF STRUCTURE FRAME No

STRUCTURAL COMPONENT Original Maximum Gauged Diminution Diminution

(Plating, Stiffener) thickness allowable P S

(mm) diminution Port Starboard mm % mm

(mm)

Operator's Signature ___ .. __ _____ . _____ . __ .. __ . __ __ __ __ Surveyor's Signature. __ ... __ ..... __ . .. __ .. __ ..... __ .. .. NOTES - See reverse

%

RESOLUTION MSC.197(80)

(adopted on 20 May 2005)

AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 88 -

NOTES TO THE REPORT TMS-DHT

I This report should be used for recording the thickness measurement of W.T.lO.T. transverse bulkheads.

2 Guidance for areas of measurement is indicated in appendix 3 of annex 10.

3 The single measurements recorded are to represent the average of mUltiple measurements.

4 The maximum allowable diminution could be stated in an attached document.

3 The single measurements recorded are to represent the average of mUltiple measurements.
4 The maximum allowable diminution could be stated in an attached document.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 89-
TM6-DHT Report on thickness measurement of miscellaneous structural members
Ship's name .... Class identity No ..... Report No ....
STRUCTURAL MEMBER SKETCH
LOCATION OF STRUCTURE
Description Original Maximum Gauged Diminution Diminution
thickness allowable P S
(rum) diminution P S mm % mm %
(rum)
Operator's Signature .... Surveyor 's Signature .... NOTES - See reverse
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 90-
NOTES TO THE REPORT TM6-DHT

RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 90-
NOTES TO THE REPORT TM6-DHT
This report should be used for recording the thickness measurement of miscellaneous structural members.
2 The single measurements recorded are to represent the average of multiple measurements.
3 The maximum allowable diminution could be stated in an attached document.
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)o.
1.
2.
3.
4.
5.
6.
- 91 -
APPENDIX 3
Thickness measurement - Double hull oil tankers
Typical transverse section of a double hull oil tanker above 150,000 dwt with indication oflongitudinal and transverse members
31
3
J2
Report on TM2-DHT
(i) and (ii)
Strength deck plating 10.
Stringer plate 11.
Sheerstrake 12.
Side shell plating 13.
Bilge plating 14.
Bottom shell plating 15.
Keel plate 16.
17.
18.
19.
CROSS TIE IN
CENTRE TANK
10
21
18

31
3
J2
Report on TM2-DHT
(i) and (ii)
Strength deck plating 10.
Stringer plate 11.
Sheerstrake 12.
Side shell plating 13.
Bilge plating 14.
Bottom shell plating 15.
Keel plate 16.
17.
18.
19.
CROSS TIE IN
CENTRE TANK
10
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Report on TM3-DHT
Deck longitudinal s 20. Longitudinal
bulkhead plating
(rernainde~)
Sheerstrake 21. Longitudinal
longitudinals bulkhead
longitudinals
Side shell 22. Inner side plating
longitudinals
Bilge longitudinals 23. Inner side
longitudinals
Bottom 24. Hopper plating
longi tudinals
Deck girders 25. Hopper
longitudinals
"
Horizontal girders 26. Inner bottom plating
in wing ballast
tanks
Bottom girders 27. Inner bottom
longi tudinals
Longitudinal 28. Topside tank plating
bulkhead top
strake
Longitudinal 29. Topside tank
bulkhead bottom longitudinals
strake
11
"
12
13
Report on TM4-DHT
30. Deck transverse -
centre tank
31. Deck transverse -
wing tank
32. Vertical web in
wing ballast tank

33. Double bottom
floor - wing tank
34. Double bottom
floor - centre tank
35. Longitudinal
bulkhead vertical
web
36. Cross ties
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wing tank
32. Vertical web in
wing ballast tank
33. Double bottom
floor - wing tank
34. Double bottom
floor - centre tank
35. Longitudinal
bulkhead vertical
web
36. Cross ties
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PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 92-
Thickness measurement - Double hull oil tankers
Typical transverse section of a double hull oil tanker up to 150,000 dwt with
indication ofiongitudinal and transverse members
e TOPSIDE TANK MULTIPLE DECK GIRDERS
o
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CENTRELINE BULKHEAD
'0
Report on TM2-DHT (i) Report on TM3-DHT Report on TM4-DHT
and (ii)
o. Strength deck 10. Deck longitudinal s 20. Longitudinal 30. Deck transverse -
plating bulkhead plating centre tank
(remainder)
I. Stringer plate II. Sheerstrake 21. Longitudinal 31. Deck transverse -
longitudinals bulkhead wing tank
longitudinals
2. Sheerstrake 12. Side shell 22. Inner side plating 32. Vertical web in
longitudinals wing ballast tank
3. Side shell plating 13. Bilge longitudinal s 23. Inner side 33. Double bottom
longitudinals floor - wing tank

longitudinals
2. Sheerstrake 12. Side shell 22. Inner side plating 32. Vertical web in
longitudinals wing ballast tank
3. Side shell plating 13. Bilge longitudinal s 23. Inner side 33. Double bottom
longitudinals floor - wing tank
4. Bilge plating 14. Bottom longitudinals 24. Hopper plating 34. Double bottom
floor - centre tank
5. Bottom shell 15. Deck girders 25. Hopper 35. Longitudinal
plating longitudinals bulkhead vertical
web
6. Keel plate 16. Horizontal girders in 26. Inner bottom 36. Cross ties
wing ballast tanks plating
17. Bottom girders 27. Inner bottom
longitudinals
18. Longitudinal 28. Topside tank
bulkhead top strake plating
19. Longitudinal 29. Topside tank
bulkhead bottom longitudinals
strake
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AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 93 -
Thickness measurement - Double hull oil tankers
Transverse section outline.
(The diagram may be used for those ships where typical sections are not applicable.)
Report on TM2-DHT Report on TM3-DHT Report on TM4-DHT
(i) and (ii)

Thickness measurement - Double hull oil tankers
Transverse section outline.
(The diagram may be used for those ships where typical sections are not applicable.)
Report on TM2-DHT Report on TM3-DHT Report on TM4-DHT
(i) and (ii)
O. Strength deck plating 10. Deck longitudinals 20. Longitudinal 30. Deck transverse - bulkhead plating centre tank
(remainder)
1. Stringer plate II. Sheerstrake 21. Longitudinal 31. Deck transverse - longitudinals bulkhead wing tank
longitudinals
2. Sheerstrake 12. Side shell 22. Inner side plating 32. Vertical web in longitudinals wing ballast tank
3. Side shell plating 13. Bilge longitudinal s 23. Inner side 33. Double bottom longitudinals floor - wing tank
4. Bilge plating 14. Bottom 24. Hopper plating 34. Double bottom longitudinals floor - centre tank
5. Bottom shell plating 15. Deck girders 25. Hopper 35. Longitudinal longitudinals bulkhead vertical
web
6. Keel plate 16. Horizontal girders 26. Inner bottom 36. Cross ties in wing ballast plating
tanks
17. Bottom girders 27. Inner bottom
longitudinal s
18. Longitudinal 28. Topside tank
bulkhead top strake plating
19. Longitudinal 29. Topside tank
bulkhead bottom longitudinals

in wing ballast plating
tanks
17. Bottom girders 27. Inner bottom
longitudinal s
18. Longitudinal 28. Topside tank
bulkhead top strake plating
19. Longitudinal 29. Topside tank
bulkhead bottom longitudinals
strake
RESOLUTION MSC.197(80)
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AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 94-
Thickness measurement - double hull oil tankers
Areas subject to close-up survey and thickness measurements - areas (1) to (5) as
defined in annex 1- thickness to be reported on TM3-DHT, TM4-DHT and
TM5-DHT, as appropriate
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RESOLUTION MSC.197(80)
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AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 95 -
Areas subject to close-up survey and thickness measurements - areas (6) to (7) as
defined in annex 1 - thickness to be reported on TM3-DHT and TM4-DHT, as
appropriate
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: Close-up survey:
area
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Areas subject to close-up survey and thickness measurements - areas (6) to (7) as defined in annex 1 - thickness to be reported on TM3-DHT and TM4-DHT, as appropriate

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RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 96-
ANNEX 11
GUIDELINES FOR TECHNICAL ASSESSMENT IN CONJUNCTION WITH THE PLANNING OF ENHANCED SURVEYS FOR OIL TANKERS
Renewal survey
1 Introduction
These Guidelines contain information and suggestions concerning technical assessments which may be of use in conjunction with the planning of renewal surveys of oil tankers. As indicated in 5.1.5, these Guidelines are a recommended tool which may be invoked at the discretion of an Administration, when considered necessary and appropriate, in

As indicated in 5.1.5, these Guidelines are a recommended tool which may be invoked at the discretion of an Administration, when considered necessary and appropriate, in conjunction with the preparation of the required survey programme.

2 Purpose and principles

2.1 Purpose

The technical assessments described in these Guidelines should assist in identifying critical structural areas, nominating suspect areas and in focusing attention on structural elements or areas of structural elements which may be particularly susceptible to, or evidence a history of, wastage or damage. This information may be useful in nominating locations, areas and tanks for thickness measurement, close-up survey and tank testing.

2.2 Minimum requirements

These Guidelines may not be used to reduce the requirements of annexes I, 2 and 3 for close-up survey, thickness measurement and tank testing, respectively, which are, in all cases, to be complied with as a minimum.

2.3 Timing

As with other aspects of survey planning, the technical assessments described in these Guidelines should be completed out by the owner or operator in co-operation with the

cases, to be complied with as a minimum.

2.3 Timing

As with other aspects of survey planning, the technical assessments described in these Guidelines should be completed out by the owner or operator in co-operation with the Administration well in advance of the commencement of the renewal survey, i.e., prior to commencing the survey and normally at least 12 to 15 months before the survey's completion due date.

2.4 Aspects to be considered

Technical assessments, which may include quantitative or qualitative evaluation of relative risks of possible deterioration, of the following aspects of a particular ship may be used as a basis for the nomination of tanks and areas for survey of:

.1 design features such as stress levels on various structural elements, design details and extent of use of high-tensile steel (HTS);

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.2 former history with respect to corrosion, cracking, buckling, indents and

.2 former history with respect to corrosion, cracking, buckling, indents and repairs for the particular ship as well as similar vessels, where available; and

.3 information with respect to types of cargo carried, use of different tanks for cargolballast, protection of tanks and condition of coating, if any.

Technical assessments of the relative risks of susceptibility to damage or deterioration of various structural elements and areas should be judged and decided on the basis of recognized principles and practices, such as may be found in references I and 2.

3 Technical assessment

3.1 General

3.1.1 There are three basic types of possible failure which may be the subject of a technical assessment in connection with the planning of surveys: corrosion, cracks and buckling. Contact damages are not normally covered by the survey plan since indents are usually noted in memoranda and assumed to be dealt with as a normal routine by surveyors.

buckling. Contact damages are not normally covered by the survey plan since indents are usually noted in memoranda and assumed to be dealt with as a normal routine by surveyors.

3.1.2 Technical assessments performed in conjunction with the survey planning process should, in principle, be as shown schematically in figure 1 which depicts how technical assessments can be carried out in conjunction with the survey planning process. The approach is basically an evaluation of the risk, based on the knowledge and experience related to design and corrosion.

3.1.3 The design should be considered with respect to structural details which may be susceptible to buckling or cracking as a result of vibration, high stress levels or fatigue.

3.1.4 Corrosion is related to the ageing process and is closely connected with the quality of corrosion protection at newbuilding and subsequent maintenance during the service life. Corrosion may also lead to cracking and/or buckling.

3.2 Methods

3.2.1 Design details

3.2.1.1 Damage experience related to the ship in question and similar ships, where available, are the main source of information to be used in the process of planning. In

3.2 Methods
3.2.1 Design details
3.2.1.1 Damage experience related to the ship in question and similar ships, where available, are the main source of information to be used in the process of planning. In addition, a selection of structural details from the design drawings should be included. Typical damage experience to be considered will consist of:
.1.1 number, extent, location and frequency of cracks; and
.1.2 location of buckles.
3.2.1.2 This information may be found in the survey reports and/or the owner's files, including the results of the owner's own inspections. The defects should be analysed, noted and marked on sketches.
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AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 98-
3.2.1.3 In addition, general experience should be utilized. For example, reference should be made to reference I, which contains a catalogue of typical damages and proposed repair methods for various tanker structural details.
3.2.1.4 Such figures should be used together with a review of the main drawings, in

should be made to reference I, which contains a catalogue of typical damages and proposed repair methods for various tanker structural details.

3.2.1.4 Such figures should be used together with a review of the main drawings, in order to compare with the actual structure and search for similar details which may be susceptible to damage. An example is shown in figure 2. In particular, chapter 3 of reference 1 deals with various aspects specific to double hull tankers , such as stress concentration locations , misalignment during construction, corrosion trends, fatigue considerations and areas requiring special attention, which should be considered in working out the survey planning.

3.2.1.5 The review of the main structural drawings , in addition to using the above-mentioned figures , should include checking for typical design details where cracking has been experienced. The factors contributing to damage should be carefully considered.

3.2.1.6 The use of HTS is an important factor. Details showing good service experience where ordinary, mild steel has been used may be more susceptible to damage

considered.
3.2.1.6 The use of HTS is an important factor. Details showing good service experience where ordinary, mild steel has been used may be more susceptible to damage when HTS, and its higher associated stresses, are utilized. There is extensive and, in general, good experience, with the use of HTS for longitudinal material in deck and bottom structures. Experience in other locations, where the dynamic stresses may be higher, is less favourable, e.g. side structures .
3.2.1. 7 In this respect, stress calculations of typical and important components and details , in accordance with relevant methods , may prove useful and should be considered.
3.2.1.8 The selected areas of the structure identified during this process should be recorded and marked on the structural drawings to be included in the survey programme .
3.2.2 Corrosion
3.2.2. I In order to evaluate relative corroSIOn risks, the following information IS generally to be considered:
.1.1 usage of tanks and spaces;
· 1.2 condition of coatings;
· 1.3 condition of anodes;
· I .4 cleaning procedures;
· 1.5 previous corrosion damage;
· 1.6 ballast use and time for cargo tanks;

generally to be considered:
.1.1 usage of tanks and spaces;
· 1.2 condition of coatings;
· 1.3 condition of anodes;
· I .4 cleaning procedures;
· 1.5 previous corrosion damage;
· 1.6 ballast use and time for cargo tanks;
.1. 7 corrosion risk scheme (see reference 2, table 2.1); and
.1. 8 location of heated tanks.
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PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 99-
3.2.2.2 Reference 2 gives definitive examples which can be used for judging and describing coating condition, using typical pictures of conditions.
3.2.2.3 The evaluation of corrosion risks should be based on information in reference 2, together with the age of the ship and relevant information on the anticipated condition as derived from the information collected in order to prepare the survey programme .
3.2.2.4 The various tanks and spaces should be listed with the corrosion risks nominated accordingly. Special attention should be given to the areas where the double

programme .
3.2.2.4 The various tanks and spaces should be listed with the corrosion risks nominated accordingly. Special attention should be given to the areas where the double hull tanker is particularly exposed to corrosion. To this end, the specific aspects addressing corrosion in double hull tankers indicated in 3.4 (Corrosion trends) of reference I should be taken into account.
3.2.3 Locations for close-up survey and thickness measurement
3.2.3.1 On the basis of the table of corrosion risks and the evaluation of design experience, the locations for initial close-up survey and thickness measurement (sections) may be nominated.
3.2.3.2 The sections subject to thickness measurement should normally be nominated in tanks and spaces where corrosion risk is judged to be the highest.
3.2.3.3 The nomination of tanks and spaces for close-up survey should, initially , be based on highest corrosion risk and should always include ballast tanks. The principle for the selection should be that the extent is increased by age or where information is insufficient or umeliable.
REFERENCES
Tanker Structure Co-operative Forum (TSCF), "Guidelines for the Inspection and

for the selection should be that the extent is increased by age or where information is insufficient or umeliable.

REFERENCES

Tanker Structure Co-operative Forum (TSCF), "Guidelines for the Inspection and Maintenance of Double Hull Tanker Structures , 1995"

2 Tanker Structure Co-operative Forum (TSCF), "Guidance Manual for Tanker Structures, 1997"

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Design Related Risk

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Hull Damage

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AnalyseHuU

Damage for

Similar Ship.

Where Available

Hull Damage : : I

General Experience

Present Areas where Damage has been found and Risks considered high, Mark Sketches or Drawings

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Collection of Information

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Coating condition

Anodes Condition

Usage of Tanks

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Corrosion Damage Similar Ship where Available

Location for Thickness Measurement and Close-Up Survey

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Corrosion Damage
This Ship
Corrosion Damage
Similar Ship where
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Location for Thickness Measurement and Close-Up Survey
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Survey Programme
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Survey
FIGURE 1:. TECHNICAL ASSESSMENT AND THE SURYEY PLANNING PROCESS
RESOLUTION MSC.197(80)
(adopted on 20 May 2005)
AMENDMENTS TO THE GUIDELINES ON THE ENHANCED
PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)LOCATION:
EXAMPLE No.1:
- 101 -
Connection of longitudinals to transverse webs
Web and flat bar fractures at cut·outs for longitudinal stiffener
connections
TYPICAL DAMAGE PROPOSED REPAIR
*FRACTURE
FULL COLIAR IF FRACTURES IN
WEB PLATE ARE SMALL AND
ARE REPAIRED BY WELDING
SHELL PIATING
OR LONGITUDINAL
BULKHEAD
FRACTURED
WELDIWEB
VIEW A-A
SIDE SHELL OR
BULKHEAD
LONGITUDINAL
A
WEB PIATING
WEB FLAT BAR
STIFFENER
LUG
VIEW A·A
NOTE *ONE OR MORE FRACTURES
MAY OCCUR
WEB AND FLAT BAR CROPPED
AND PART RENEWED OR
ALTERNATIVELY WELDED
FACTORS CONTRIBUTING TO DAMAGE

VIEW A-A
SIDE SHELL OR
BULKHEAD
LONGITUDINAL
A
WEB PlATING
WEB FLAT BAR
STIFFENER
LUG
VIEW A·A
NOTE *ONE OR MORE FRACTURES
MAY OCCUR
WEB AND FLAT BAR CROPPED
AND PART RENEWED OR
ALTERNATIVELY WELDED
FACTORS CONTRIBUTING TO DAMAGE
1. Asynunetrical colUlection of flat bar stiffener resulting in high peak stresses at the heel of
the stiffener under fatigue loading.
2. Insufficient area of connection of loogitudinal to web plate.
3. Defective weld at return around the plate thickness.
4. High localised corrosion at areas of stress concentratioo such BS flat bar stiffener connections,
comera of cut-<lut for the longitudinal and connection of web to shell at cut-<luts.
5 High stress in the web of the transverse.
6. Dynamic sea way load/ship motions.
RGURE
1
TANKER STRUCTURE CO-OPERATIVE FORUM
SUBJECT: CATALOGUE OF STRUCTURAL DETAILS
FIGURE 2: TYPICAL DAMAGE AND REPAIR EXAMPLE
(REPRODUCED FROM REF. 2)
FIGURE
1
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PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
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ANNEX 12

FIGURE
1
RESOLUTION MSC.197(80)
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PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS
AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 102-
ANNEX 12
CRITERIA FOR LONGITUDINAL STRENGTH OF HULL GIRDER
FOR OIL TANKERS
1 General
1.1 These criteria should be used for the evaluation of the longitudinal strength of the ship's hull girder as required by 8.1.2.
1.2 In order that the ship's longitudinal strength to be evaluated can be recognized as valid, fillet welding between longitudinal internal members and hull envelopes should be in sound condition so as to keep the integrity of longitudinal internal members with hull envelopes.
2 Evaluation oflongitudinal strength
On oil tankers of 130 m in length and upwards and over 10 years of age, the longitudinal strength of the ship's hull girder should be evaluated in compliauce with the requirements of this annex on the basis of the thickness measured, renewed or reinforced, as appropriate, during the renewal survey of the Cargo Ship Safety Construction Certificate

of this annex on the basis of the thickness measured, renewed or reinforced, as appropriate, during the renewal survey of the Cargo Ship Safety Construction Certificate or Cargo Ship Safety Certificate (SC renewal survey). The condition of the hull girder for longitudinal strength evaluation should be determined in accordance with the methods specified in appendix 3.

2.1 Calculation of transverse sectional areas of deck and bottom flanges of hull girder

2.1.1 The transverse sectional areas of deck flange (deck plating and deck longitudinals) and bottom flange (bottom shell plating and bottom longitudinals) of the ship's hull girder should be calculated by using the thickness measured, renewed or reinforced, as appropriate, during the SC renewal survey.

2.1.2 If the diminution of sectional areas of either deck or bottom flange exceeds 10% of their respective as-built area (i.e., original sectional area when the ship was built), either one ofthe following measures should be taken:

.1 to renew or reinforce the deck or bottom flanges so that the actual sectional area is not less than 90% of the as-built area; or

either one ofthe following measures should be taken:
.1 to renew or reinforce the deck or bottom flanges so that the actual sectional area is not less than 90% of the as-built area; or
.2 to calculate the actual section of moduli (Zact) of transverse section of the ship's hull girder by applying the calculation method specified in appendix I, by using the thickness measured, renewed or reinforced, as appropriate, during the SC renewal survey.
2.2 Requirements for transverse section modulus of hull girder
The actual section moduli of the transverse section of the ship's hull girder, calculated in accordance with 2.1.2.2, should satisfy either of the following provisions, as applicable:
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.1 for ships constructed on or after 1 July 2002, the actual section moduli (Zaet) of the transverse section of the ship's hull girder calculated in accordance with the requirements of 2.1.2.2 should be not less than the

.1 for ships constructed on or after 1 July 2002, the actual section moduli (Zaet) of the transverse section of the ship's hull girder calculated in accordance with the requirements of 2.1.2.2 should be not less than the diminution limits determined by the Administration, taking into account the recommendations adopted by the Organization; or

.2 for ships constructed before 1 July 2002, the actual section moduli (Zoet) of the transverse section of the ship's hull girder calculated in accordance with the requirements of 2.1.2.2 should meet the criteria for minimum section modulus for ships in service required by the Administration or recognized classification society, provided that in no case Zaet should be less than the diminution limit of the minimum section modulus (Zme) as specified in appendix 2.

Appendix 1

Calculation criteria of section moduli of midship section of hull girder

When calculating the transverse section modulus of the ship's hull girder, the sectional area of all continuous longitudinal strength members should be taken into account.

2 Large openings, i.e. openings exceeding 2.5 m in length or 1.2 m in breadth, and

sectional area of all continuous longitudinal strength members should be taken into account.

2 Large openings, i.e. openings exceeding 2.5 m in length or 1.2 m in breadth, and scallops, where scallop welding is applied, are always to be deducted from the sectional areas used in the section modulus calculation.

3 Smaller openings (manholes, lightening holes, single scallops in way of seams, etc.) need not be deducted, provided that the sum of their breadths or shadow area breadths in one transverse section does not reduce the section modulus at deck or bottom by more than 3% and provided that he height of lightening holes, draining holes and single scallops in longitudinals or longitudinal girders does not exceed 25% of the web depth, for scallops maximum 75 mm.

4 A deduction-free sum of smaller opening breadths in one transverse section in the bottom or deck area of 0.06 (8 - Lb) (where 8 = breadth of ship, Lb = total breadth of large openings) may be considered equivalent to the above reduction in sectional modulus.

5 The shadow area will be obtained by drawing two tangent lines with an opening angle of30°.

large openings) may be considered equivalent to the above reduction in sectional modulus.
5 The shadow area will be obtained by drawing two tangent lines with an opening angle of30°.
6 The deck modulus is related to the moulded deck line at side.
7 The bottom modulus is related to the baseline.
8 Continuous trunks and longitudinal hatch coamings should be included in the longitudinal sectional area provided they are effectively supported by longitudinal bulkheads or deep girders. The deck modulus is then to be calculated by dividing the moment of inertia by the following distance, provided this is greater than the distance to the deck line at side:
y, =y (0.9+0.2 ~)
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where:
y = distance from neutral axis to top of continuous strength member;
x = distance from top of continuous strength member to centreline of the ship;
x and y to be measured to the point giving the largest value of Yt .

where:
y = distance from neutral axis to top of continuous strength member;
x = distance from top of continuous strength member to centreline of the ship;
x and y to be measured to the point giving the largest value of Yt .
9 Longitudinal girders between multi-hatchways will be considered by special calculations.
Appendix 2
Diminution limit of minimum longitudinal strength of ships in service
1 The diminution limit of the minimum section modulus (Zm,) of oil tankers in service is given by the following formula:
where:
L = Length of ship. L is the distance, in metres, on the summer load waterline from the fore side of stem to the after side of the rudder post, or the centre of the rudder stock if there is no rudder post. L should not be less than 96%, and need not be greater than 97%, of the extreme length on the summer load waterline. In ships with unusual stem and bow arrangement, the length L may be specially considered.
B = Greatest moulded breadth in metres.
Cb = Moulded block coefficient at draught d corresponding to summer load waterline , based on Land B. Cb should not be taken less than 0.6.
Cb = moulded displacement(m 3) at draught d LBd
C = 0.9cn

B = Greatest moulded breadth in metres.
Cb = Moulded block coefficient at draught d corresponding to summer load waterline , based on Land B. Cb should not be taken less than 0.6.
Cb = moulded displacement(m 3) at draught d LBd
C = 0.9cn
Cn 10.75 - (300-L f 100 for 130 m::: L::: 300 m
Cn = 10.75 for 300 m < L < 350 m
Cn 10.75 - (L- 350 f 150 for 350 m::: L ::: 500 m
k = material factor, e.g.
k = 1.0 for mild steel with yield stress of 235 N/mm2 and over
k = 0.78 for high-tensile steel with yield stress of315 N/mm2 and over
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k = 0.72 for high-tensile steel with yield stress of355 N/mm 2 and over.
2 Scantlings of all continuous longitudinal members of the ship's hull girder based on the section modulus requirement in I above should be maintained within O.4L amidships. However, in special cases, based on consideration of type of ship, hull form and loading conditions, the scantlings may be gradually reduced towards the end of O.4L

amidships. However, in special cases, based on consideration of type of ship, hull form and loading conditions, the scantlings may be gradually reduced towards the end of O.4L part, bearing in mind the desire not to inhibit the ship's loading flexibility.

3 However, the above standard may not be applicable to ships of unusual type or design, e.g. for ships of unusual main proportions andlor weight distributions.

Appendix 3

Sampling method of thickness measurements for longitudinal strength evaluation and repair methods

1 Extent of longitudinal strength evaluation

Longitudinal strength should be evaluated within O.4L amidships for the extent ofthe hull girder length that contains tanks therein and within 0.5L amidships for adjacent tanks which may extend beyond O.4L amidships, where tanks means ballast tanks and cargo tanks.

2 Sampling method of thickness measurement

2.1 Pursuant to the requirements of section 2.5, transverse sections should be chosen such that thickness measurements can be taken for as many different tanks in corrosive environments as possible, e.g. ballast tanks sharing a common plane boundary with cargo

such that thickness measurements can be taken for as many different tanks in corrosive environments as possible, e.g. ballast tanks sharing a common plane boundary with cargo tanks fitted with heating coils, other ballast tanks, cargo tanks permitted to be filled with seawater and other cargo tanks. Ballast tanks sharing a common plane boundary with cargo tanks fitted with heating coils and cargo tanks permitted to be filled with seawater should be selected where present.

2.2 The minimum number of transverse sections to be sampled should be in accordance with annex 2. The transverse sections should be located where the largest thickness reductions are suspected to occur or are revealed from deck and bottom plating measurements prescribed in 2.3 and should be clear of areas which have been locally renewed or reinforced.

2.3 At least two points should be measured on each deck plate andlor bottom shell plate required to be measured within the cargo area in accordance with the requirements of annex 2.

2.4 Within 0.1D (where D is the ship's moulded depth) of the deck and bottom at each transverse section to be measured in accordance with the requirements of annex 2,

of annex 2.
2.4 Within 0.1D (where D is the ship's moulded depth) of the deck and bottom at each transverse section to be measured in accordance with the requirements of annex 2, every longitudinal and girder should be measured on the web and face plate, and every plate should be measured at one point between longitudinals.
2.5 For longitudinal members other than those specified in 2.4 to be measured at each transverse section in accordance with the requirements of annex 2, every longitudinal and girder should be measured on the web and face plate, and every plate should be measured at least in one point per strake.
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2.6 The thickness of each component should be determined by averaging all of the measurements taken in way of the transverse section on each component.
3 Additional measurements where the longitudinal strength is deficient
3.1 Where one or more of the transverse sections are found to be deficient in respect

measurements taken in way of the transverse section on each component.
3 Additional measurements where the longitudinal strength is deficient
3.1 Where one or more of the transverse sections are found to be deficient in respect of the longitudinal strength requirements given in this annex, the number of transverse sections for thickness measurement should be increased such that each tank within the O.5L amidships region has been sampled. Tank spaces that are partially within, but extend beyond, the O.5L region, should be sampled.
3.2 Additional thickness measurements should also be performed on one transverse section forward and one aft of each repaired area to the extent necessary to ensure that the areas bordering the repaired section also comply with the requirements of the Guidelines.
4 Effective repair methods
4.1 The extent of renewal or reinforcement carried out to comply with this annex should be in accordance with 4.2.
4.2 The minimum continuous length of a renewed or reinforced structural member should be not less than twice the spacing of the primary members in way. In addition, the

should be in accordance with 4.2.

4.2 The minimum continuous length of a renewed or reinforced structural member should be not less than twice the spacing of the primary members in way. In addition, the thickness diminution in way of the butt joint of each joining member forward and aft of the replaced member (plates, stiffeners, girder webs and flanges, etc.) should not be within the substantial corrosion range (75% of the allowable diminution associated with each particular member). Where differences in thickness at the butt joint exceed 15% of the lower thickness, a transition taper should be provided.

4.3 Alternative repair methods involving the fitting of straps or structural member modification should be subject to special consideration. In considering the fitting of straps, it should be limited to the following conditions:

.1 to restore and/or increase longitudinal strength;

.2 the thickness diminution of the deck or bottom plating to be reinforced should not be within the substantial corrosion range (75% of the allowable diminution associated with the deck plating);

.3 the alignment and arrangement, including the termination of the straps, is

should not be within the substantial corrosion range (75% of the allowable diminution associated with the deck plating);
.3 the alignment and arrangement, including the termination of the straps, is in accordance with a standard recognized by the Administration;
.4 the straps are continuous over the entire O.5L amidships length; and
.5 continuous fillet welding and full penetration welds are used at butt welding and, depending on the width of the strap, slot welds. The welding procedures applied should be acceptable to the Administration.
4.4 The existing structure adjacent to replacement areas and in conjunction with the fitted straps, etc., should be capable of withstanding the applied loads, taking into account the buckling resistance and the condition of welds between the longitudinal members and hull envelope plating."

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Part B
GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF OIL TANKERS OTHER THAN DOUBLE HULL OIL TANKERS

PROGRAMME OF INSPECTIONS DURING SURVEYS OF BULK CARRIERS AND OIL TANKERS (RESOLUTION A.744(18), AS AMENDED)- 107 -

Part B

GUIDELINES ON THE ENHANCED PROGRAMME OF INSPECTIONS DURING SURVEYS OF OIL TANKERS OTHER THAN DOUBLE HULL OIL TANKERS

44 The text of the new Part B is the text of existing Annex B subject to the following amendments.

45 The existing paragraph 1.1.1 is replaced with the following:

"1.1.1 The Guidelines should apply to self-propelled oil tankers of 500 gross tonnage and above other than double hull oil tankers, as defined in 1.2.1 of Part A of Annex B."

46 Paragraph 1.1.2 is deleted and paragraphs 1.1.3 and 1.1.4 are renumbered as paragraphs 1.1.2 and 1.1.3.

47 At the end of the new paragraph 1.1.2 (existing paragraph 1.1.3), the following sentence is added:

"The surveys should be carried out during the surveys prescribed by regulation 1/10 of the 1974 SOLAS Convention, as amended."

48 In paragraph 1.2.ll, the reference "II-2/3.32" is replaced with "II-2/3.6".

49 In paragraph 1.2.12, the words "Intermediated enhanced survey is an enhanced survey" are replaced by "Intermediate survey is a survey".

48 In paragraph 1.2.ll, the reference "II-2/3.32" is replaced with "II-2/3.6".
49 In paragraph 1.2.12, the words "Intermediated enhanced survey is an enhanced survey" are replaced by "Intermediate survey is a survey".
50 In paragraph 1.2.l3, the words "condition of classification" are replaced by "condition of classification or recommendation".
51 A new paragraph 1.2.14 is added as follows:
"1.2.14 Specially considered means sufficient close-up inspection and thickness measurements are taken to confirm the actual average condition of the structure under coating."
52 A new subitem ".6 items in 3.3." is added in paragraph 1.3.1.
53 The word "significant" in paragraph 1.3.2 is deleted.
54 The following new paragraph 1.4 is added:
"1.4 Surveyors
For tankers of 20,000 tons deadweight and above, two surveyors should jointly carry out the first scheduled renewal survey after the tanker passes 10 years of age, and all subsequent renewal surveys and intermediate surveys. If the surveys are carried out by a recognized organization, the surveyors should be exclusively employed by such recognized organizations."

subsequent renewal surveys and intermediate surveys. If the surveys are carried out by a recognized organization, the surveyors should be exclusively employed by such recognized organizations."

55 The words "thickness measurement and" in paragraph 2.1.2 are deleted.

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56 The words "Cargo Ship Safety Construction" are inserted between the words "the" and "Certificate" in paragraphs 2.1.3 and 2.2.1.

57 The words "issued to oil tankers referred to in paragraph 1.1.1 and/or the International Oil Pollution Prevention Certificate, as appropriate, issued to oil tankers referred to in paragraph 1.1.2" in paragraph 2.2.3 are deleted.

58 Paragraph 2.2.4 is deleted.

59 The references to a footnote regarding "specially considered" in paragraphs 2.4.4 and 2.5.4 are deleted.

60 The words "or as specified ill planning document as described III annex 6" III paragraph 2.5.2 are deleted.

61 In paragraph 3.1, the word "annual" is inserted before the word "survey".

2.5.4 are deleted.
60 The words "or as specified ill planning document as described III annex 6" III paragraph 2.5.2 are deleted.
61 In paragraph 3.1, the word "annual" is inserted before the word "survey".
62 The words ", oily ballast" are deleted in paragraph 3.3.3.
63 In paragraph 3.5.3, the words "a cargo tank" are replaced by "a cargo or fuel tank".
64 The words "For tanks used for salt water ballast including combined cargolballast tanks" in paragraph 4.3.3 are replaced by "For ballast tanks".
65 Existing paragraph 5.1.1 is replaced by the following:
"5.1.1 A specific survey programme should be worked out in advance of the renewal survey by the owner in co-operation with the Administration. The survey programme should be in a written format based on the information in Annex 6A. The survey should not commence until the survey programme has been agreed.
5.1.1.1 Prior to the development of the survey programme, the survey planning questionnaire should be completed by the owner based on the information set out in Annex 6B, and forwarded to the Administration."
66 The existing paragraph 5.1.2 is replaced with the following:

questionnaire should be completed by the owner based on the information set out in Annex 6B, and forwarded to the Administration."

66 The existing paragraph 5.1.2 is replaced with the following:

"5.1.2 In developing the survey programme, the following documentation should be collected and consulted with a view to selecting tanks, areas, and structural elements to be examined:

.1 survey status and basic ship information;

.2 documentation on board, as described in 6.2 and 6.3;

.3 main structural plans of cargo and ballast tanks (scantlings drawings), including information regarding use of high-tensile steels (HTS);

.4 Condition Evaluation Report, according to annex 9;

.5 relevant previous damage and repair history;

.6 relevant previous survey and inspection reports from both the recognized organization and the owner;

.7 cargo and ballast history for the last 3 years, including carriage of cargo under heated conditions;

.6 relevant previous survey and inspection reports from both the recognized organization and the owner;
.7 cargo and ballast history for the last 3 years, including carriage of cargo under heated conditions;
.8 details of the inert gas plant and tank cleaning procedures;
.9 information and other relevant data regarding conversion or modification of the ship's cargo and ballast tanks since the time of construction;
.10 description and history of the coating and corrosion protection system (including anodes and previous class notations), if any;
.11 inspections of the Owner's personnel during the last 3 years with reference to structural deterioration in general, leakages in tank boundaries and piping and condition of the coating and corrosion protection system (including anodes), if any. A guidance for reporting is shown in annex 5;
.12 information regarding the relevant maintenance level during operation including port state control reports of inspection containing hUll related deficiencies, Safety Management System non-conformities relating to hull maintenance, including the associated corrective action(s); and

including port state control reports of inspection containing hUll related deficiencies, Safety Management System non-conformities relating to hull maintenance, including the associated corrective action(s); and

.13 any other information that will help identifY suspect areas and critical structural areas."

67 The existing paragraph 5.1.3 is replaced with the following:

"5.1.3 The submitted survey programme should account for and comply, as a minimum , with the requirements of 2.6 and annexes 1, 2 and 3 for close-up survey, thickness measurement and tank testing, respectively, and should include relevant information including at least:

.1 basic ship information and particulars;

.2 main structural plans of cargo and ballast tanks (scantling drawings), including information regarding use of high tensile steels (HTS);

.3 arrangement of tanks;

.4 list of tanks with information on their use, extent of coatings and corrosion protection systems;

.5 conditions for survey (e.g., information regarding tank cleaning, gas freeing, ventilation, lighting, etc.);

.6 provisions and methods for access to structures;

.7 equipment for surveys;

protection systems;
.5 conditions for survey (e.g., information regarding tank cleaning, gas freeing, ventilation, lighting, etc.);
.6 provisions and methods for access to structures;
.7 equipment for surveys;
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.8 identification of tanks and areas for close-up survey (see 2.4);
.9 identification of areas and sections for thickness measurement (see 2.5);
.10 identification of tanks for tank testing (see 2.6);
.11 identification of the thickness measurement company;
.12 damage experience related to the ship in question; and
.13 critical structural areas and suspect areas, where relevant."

68 Paragraph 5.1.4 is deleted, and paragraphs 5.1.5 and 5.1.6 are renumbered as 5.1.4 and 5.1.5.

69 The following new paragraphs 5.2.1.1 to 5.2.1.3 are added after existing paragraph 5.2.1:

"5.2.1.1 In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access should be agreed between the owner and the Administration.

"5.2.1.1 In order to enable the attending surveyors to carry out the survey, provisions for proper and safe access should be agreed between the owner and the Administration.
5.2.1.2 Details of the means of access should be provided in the survey planoing questionoaire.
5.2.1.3 In cases where the provisions of safety and required access are judged by the attending surveyors not to be adequate, the survey of the spaces involved should not proceed."
70 Existing paragraphs 5.2.2 to 5.2.4 are replaced by the following:
"5.2.2 Tanks and spaces should be safe for access. Tanks and spaces should be gas free and properly ventilated. Prior to entering tank, void or enclosed space, it should be verified that the atmosphere in that space is free from hazardous gas and contains sufficient oxygen.
5.2.3 Tanks and spaces should be sufficiently clean and free from water, scale, dirt, oil residues, sediments, etc., to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating. In particular this applies to areas which are subject to thickness measurements.

residues, sediments, etc., to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating. In particular this applies to areas which are subject to thickness measurements.
5.2.4 Sufficient illumination should be provided to reveal corrosion, deformation, fractures, damages or other structural deterioration as well as the condition of the coating."
71 The following new paragraphs 5.2.5 and 5.2.6 are added:
"5.2.5 The surveyor(s) should always be accompanied by at least one responsible person, assigned by the owner, experienced in tank and enclosed spaces inspection. In addition a backup team of at least two experienced persons should be stationed at the hatch opening of the tank or space that is being surveyed. The back-up team should continuously observe the work in the tank or space and should keep lifesaving and evacuation equipment ready for use.
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equipment ready for use.
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5.2.6 A communication system should be arranged between the survey party in the tank or space being examined, the responsible officer on deck and, as the case may be, the navigation bridge. The communication arrangements should be maintained throughout the survey."
72 In existing paragraph 5.3.2, between the fourth and fifth subitems, a new subitem "portable ladders" is inserted.
73 The six subitems in paragraph 5.3.2 are numbered from ".1" to ".6" and the five subitems in paragraph 5.4.2 are numbered from ".1" to ".5".
74 New paragraphs 5.4.3 to 5.4.5 are added as follows:
"5.4.3 Explosimeter, oxygen-meter, breathing apparatus, lifelines, riding belts with rope and hook and whistles together with instructions and guidance on their use should be made available during the survey. A safety checklist should be provided.
5.4.4 Adequate and safe lighting should be provided for the safe and efficient conduct of the survey.

made available during the survey. A safety checklist should be provided.
5.4.4 Adequate and safe lighting should be provided for the safe and efficient conduct of the survey.
5.4.5 Adequate protective clothing should be made available and used (e.g., safety helmet, gloves, safety shoes, etc.) during the survey."
75 The existing paragraph 5.5.3 is deleted and the existing paragraph 5.5.4 is renumbered as 5.5.3.
76 The following new paragraphs 5.5.4 to 5.5.7 are added:
"5.5.4 When rafts or boats will be used for close-up survey, the following conditions should be observed:
.1 only rough duty, inflatable rafts or boats, having satisfactory residual buoyancy and stability even if one chamber is ruptured, should be used;
.2 the boat or raft should be tethered to the access ladder and an additional person should be stationed down the access ladder with a clear view of the boat or raft;
.3 appropriate lifejackets should be available for all participants;
.4 the surface of water in the tank should be calm (under all foreseeable conditions the expected rise of water within the tank should not exceed 0.25 m) and the water level either stationary or falling. On no

.4 the surface of water in the tank should be calm (under all foreseeable conditions the expected rise of water within the tank should not exceed 0.25 m) and the water level either stationary or falling. On no account should the level of the water be rising while the boat or raft is in use;
.5 the tank or space must contain clean ballast water only. Even a thin sheen of oil on the water is not acceptable;
.6 at no time should the water level be allowed to be within I m of the deepest under-deck web face flat so that the survey team is not isolated from a direct escape route to the tank hatch. Filling to levels above the deck transverses should only be contemplated if a deck access manhole is fitted and open in the bay being examined, so that an escape route for the
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survey party is available at all times. Other effective means of escape to the deck may be considered;
.7 if the tanks (or spaces) are connected by a common venting system , or

survey party is available at all times. Other effective means of escape to the deck may be considered;

.7 if the tanks (or spaces) are connected by a common venting system , or Inert Gas system, the tank in which the boat or raft should be used should be isolated to prevent a transfer of gas from other tanks (or spaces).

5.5.5 Rafts or boats alone may be allowed for inspection of the under-deck areas for tanks or spaces if the depth of the webs is 1.5 m or less.

5.5.6 If the depth of the webs is more than 1.5 m, rafts or boats alone may be allowed only:

.1 when the coating of the under-deck structure is in GOOD condition and there is no evidence of wastage; or

.2 if a permanent means of access is provided in each bay to allow safe entry and exit. This means of access should be direct from the deck via a vertical ladder with a small platform fitted approximately 2 m below the deck. Other effective means of escape to the deck may be considered.

If neither of the above conditions are met, then staging or other equivalent means should be provided for the survey ofthe under-deck areas.

deck. Other effective means of escape to the deck may be considered.
If neither of the above conditions are met, then staging or other equivalent means should be provided for the survey ofthe under-deck areas.
5.5.7 The use of rafts or boats alone in 5.5.5 and 5.5.6 does not preclude the use of boats or rafts to move about within a tank during a survey."
77 The following new section 5.6 is added:
"5.6 Survey planning meeting
5.6. I Proper preparation and close co-operation between the attending surveyor(s) and the owner's representatives on board prior to and during the survey are an essential part in the safe and efficient conduct of the survey. During the survey, on board safety meetings should be held regularly.
5.6.2 Prior to commencement of any part of the renewal and intermediate survey, a survey planning meeting should be held between the attending surveyor(s), the owner's representative in attendance , the thickness measurement company operator (as applicable) and the master of the ship for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the

and the master of the ship for the purpose to ascertain that all the arrangements envisaged in the survey programme are in place, so as to ensure the safe and efficient conduct of the survey work to be carried out.

5.6.3 The following is an indicative list of items that should be addressed III the meeting:

.I schedule of the vessel (i.e. the voyage , docking and undocking manoeuvres, periods alongside , cargo and ballast operations, etc.);

.2 provisions and arrangements for thicknes s measurements (i.e. access, cleaning /de-scaling , illumination, ventilation , personal safety);

.3 extent of the thickness measurements;

.4 acceptance criteria (refer to the list of minimum thicknesses);

.5 extent of close-up survey and thickness measurement considering the coating condition and suspect areas/areas of substantial corrosion;

.6 execution of thickness measurements;

.7 taking representative readings III general and where uneven corrosion/pitting is found;

coating condition and suspect areas/areas of substantial corrosion;
.6 execution of thickness measurements;
.7 taking representative readings III general and where uneven
corrosion/pitting is found;
.8 mapping of areas of substantial corrosion;
.9 communication between attending surveyor(s) the thickness measurement
company operator(s) and owner representative(s) concerning findings."
78 The words "supply and maintain on-board" in paragraph 6.1.1 are replaced with "obtain, supply and maintain on board the ship".
79 Paragraph 6.2.1.4 is deleted.
80 The word "inspection" in paragraph 6.4 is replaced with "survey".
81 The word "extend" in paragraph 7.1.3 is replaced with "extent".
82 The following new paragraph 8.2.2 is added after existing paragraph 8.2.1:
"8.2.2 When a survey is split between different survey stations, a report should be made for each portion of the survey. A list of items examined and/or tested (pressure testing, thickness measurements, etc.) and an indication of whether the item has been credited, should be made available to the next attending surveyor(s), prior to continuing or completing the survey."

thickness measurements, etc.) and an indication of whether the item has been credited, should be made available to the next attending surveyor(s), prior to continuing or completing the survey."

83 The existing paragraph 8.2.2 is renumbered as 8.2.3.

84 In the third column of annex I, the words "One web frame rings in each remaining cargo wing tank" are replaced with "A minimum of 30% of all web frame rings in each remaining cargo wing tank (see Note 1)" and the words "One deck and bottom transverse in each cargo centre tank" are replaced with "A minimum of 30% of deck and bottom transverses, including adjacent structural members in each cargo centre tank (see Note 1)". A Note 1 is added at the bottom of that annex as:

"Note I: The 30% should be rounded up to the next whole integer."

85 A new annex 6A "Survey programme" is added after existing annex 5. The text of the annex is identical to annex 6A of Part A (see amendment no.43).

86 A new annex 6B is added after the new annex 6A. The text of the annex is identical to annex 6B of Part A (see amendment no.43).

87 The existing annex 6 is deleted.

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86 A new annex 6B is added after the new annex 6A. The text of the annex is identical to annex 6B of Part A (see amendment no.43).
87 The existing annex 6 is deleted.
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88 The existing text of annex 8 is replaced by new text with the heading "Survey reporting principles". The new text is identical to the text of annex 8 of Part A (see amendment no.43).
89 Table I and table 2 in appendix 3 of annex 10 are deleted and table 3 is renumbered as table 1.
90 In paragraph I of annex 11, the words "paragraph 5.1.6 of annex B" are replaced by "5.1.5".
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